Multi-Class Annotation Errors in Causal Inference: Topology-Dependent Bias, Sensitivity, and Correction

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Abstract

Large language models are increasingly used as annotators in computational social science, yet annotation errors propagate into downstream causal estimates in ways that depend on both the confusion matrix structure and the causal DAG topology. We derive bias expressions for seven canonical topologies under K-class non-differential misclassification and show that $K{\geq}3$ introduces a structural possibility of sign reversal absent in the binary case. We prove sign preservation for confounding and collider, and verify it numerically for mediation, M-bias, and front-door; exposure and IV topologies can reverse the treatment effect sign. Under realistic LLM accuracy, sign reversal occurs in 2.1% of exposure configurations, in secondary coefficients with small effect sizes; across all 23 empirical LLM configurations (mean diagonal 0.45–0.81), sign reversal is confined exclusively to exposure, though IV reaches 52.6% under uniform Dirichlet priors. We introduce Annotation Sensitivity Values (ASVs), with proved ($K{=}2$) and numerically verified ($K{\geq}3$) monotonicity, yielding a vulnerability ranking from IV (maximally fragile) to M-bias (immune), and show that matrix-based corrections are provably ineffective under linear regression for four topologies due to Frisch–Waugh–Lovell invariance. Topology-aware analysis is necessary for valid causal inference with LLM-generated annotations.

1 Introduction

Large language models are rapidly replacing human annotators in computational social science, with researchers using LLM-generated labels for tasks ranging from sentiment classification to political stance detection (Gilardi et al., 2023; Ziems et al., 2024; Egami et al., 2023; Baumann et al., 2025). Crucially, most of these tasks are inherently multi-class: stance detection uses 3–4 categories, sentiment analysis 5, and disaster response taxonomies 10 or more. This shift promises dramatic cost reduction, but introduces a critical validity gap: annotation errors (equivalently, misclassification) from imperfect classifiers propagate into downstream causal estimates in ways that depend on both the error structure and the causal design. Baumann et al. (2025) demonstrate that across 37 annotation tasks, 31% of hypotheses yield incorrect conclusions with state-of-the-art LLMs, rising to 50% for smaller models; these errors are highly correlated across models (60% agreement when both err) (Kim et al., 2025), undermining ensemble-based mitigation. Analyzing the *structure* of annotation error, rather than simply reducing its magnitude, is essential for valid downstream inference.

The transition at $K{\geq}3$ is not gradual but structural. For $K{=}2$, the stochastic constraint forces any confusion matrix with positive diagonal to preserve coefficient signs—an algebraic property of the $2{\times}2$ matrix dimension (Brenner, 1993). This guarantee *vanishes* at $K{\geq}3$: the higher-dimensional confusion matrix admits negative effective weights on categories with opposite-sign coefficients, enabling sign reversal even under high accuracy (Veierød and Laake, 2001). Under a uniform Dirichlet prior, 83.4% of $K{=}3$ exposure confusion matrices produce sign reversal of at least one secondary coefficient; under realistic accuracy (diagonal $\geq$ 0.7), 2.1% still do. The primary treatment coefficient, however, is never reversed in any tested configuration—an empirical regularity we discuss in Section 3.

Standard classification metrics do not predict downstream causal risk: within a narrow F_1 range of 0.614–0.690 across three frontier LLMs, relative bias varies by an order of magnitude across topologies (from 4.1% to 42.1%). F_1 shows no significant rank correlation with causal bias across 56 HumAID scenarios (Spearman $\rho = -0.10$, $p = 0.45$), confirming that *topology*, not annotator quality, determines downstream risk. This fundamental asymmetry has received limited systematic treatment. Dosemeci et al. (1990) and Veierød and Laake (2001) noted the possibility of polytomous sign reversal; Brenner (1993) identified amplification for polytomous confounders; Lee and Wood-Doughty (2024) extended classifier error analysis to categorical variables and LLM classifiers, but covered only one DAG topology. We present the first unified analysis of K-class annotation error propagation across seven canonical DAG topologies, building on prior single-topology analyses (Brenner, 1993; TeBlunthuis et al., 2024). Our analytical predictions match Monte Carlo estimates within 1.6% MAPE across 161 scenarios.

We address these gaps with three contributions:

1. **Multi-class bias taxonomy.** We derive bias expressions for seven canonical DAG topologies[1] (Figure 1) under K-class non-differential misclassification in linear structural equation models, prove sign preservation for confounding and collider, verify it numerically for mediation, M-bias, and front-door, and show that exposure and IV can produce sign reversal. Binarization amplifies the gap: collapsing K=3 to binary introduces additional bias varying 154× across topologies (Appendix B.1).

2. **Annotation Sensitivity Values.** As a corollary of the bias taxonomy, we define three complementary ASV diagnostics—magnitude, significance, and sign-flip—revealing a topology-dependent vulnerability spectrum from IV (maximally fragile; 52.6% sign reversal under random Dirichlet prior, though zero reversals occur under empirical LLM matrices) to M-bias (immune, ASV $= \infty$).

3. **Topology-aware correction guidance** (a systematic benchmark, not a new correction method). We evaluate five correction methods and prove a Frisch–Waugh–Lovell invariance result: matrix-based corrections achieve exactly 0% bias reduction for four topologies where A enters as a covariate—a negative result overlooked in the annotation correction literature. PPI++ achieves 47.0% RMSE reduction but requires gold labels and is harmful for M-bias (−88%).

[1]These span the four structural roles of the annotated variable: treatment (exposure), outcome-side covariate (confounding, mediation, M-bias, front-door), selection variable (collider), and instrument (IV).

2 Related Work

Measurement error in causal inference. The foundations are established in Carroll et al. (2006), Gustafson (2004), Buonaccorsi (2010), and Bound et al. (2001), with starred-node DAGs formalized by Hernán and Cole (2009) and DAG-based correction theory developed by Kuroki and Pearl (2014). For binary variables, non-differential misclassification biases toward the null (Brenner, 1993; Ogburn and VanderWeele, 2012), with Neuhaus (1999) quantifying the attenuation and efficiency loss in binary regression under known misclassification rates; but this fails for polytomous exposures (Dosemeci et al., 1990; Jurek et al., 2005; Veierød and Laake, 2001). Edwards et al. (2021) showed dependent misclassification can bias in either direction, and Shu and Yi (2019b,a) develop weighted estimators that jointly handle mismeasured covariates and misclassified outcomes under IPW—a concern that also drives mediation analyses with mismeasured mediators (Imai et al., 2010a; Valeri and VanderWeele, 2013). All analyze a single DAG topology; ours is the first to characterize how the *same* confusion matrix produces qualitatively different bias across seven topologies.

Text classifiers and LLM annotation. Grimmer et al. (2022) and Feder et al. (2022) provide foundations for text-as-data and causal inference in NLP. Wood-Doughty et al. (2018) first applied DAG reasoning to classifier error; TeBlunthuis et al. (2024) gave the most complete binary treatment (K=2, four DAGs); Lee and Wood-Doughty (2024) extended to categorical LLM classifiers but

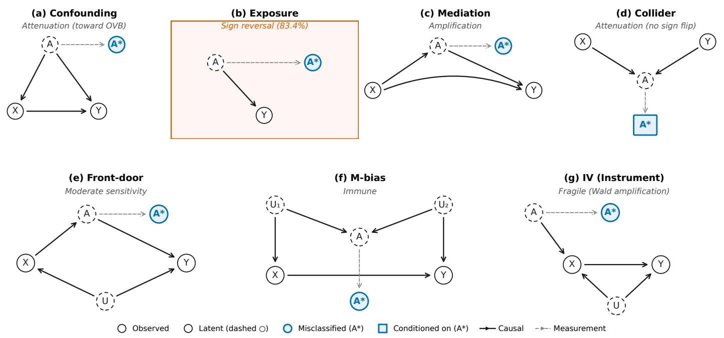

Figure 1: Seven canonical DAG topologies with starred-node annotation (Hernán and Cole, 2009). A (latent) $\rightarrow A^*$ (observed) via a dashed measurement edge; the causal role of A varies: treatment (exposure), covariate (confounding, mediation, M-bias, front-door), selection (collider), instrument (IV).

covered only confounding. On annotation quality, Gilardi et al. (2023) and Ziems et al. (2024) document the rapid adoption of LLM annotators, Baumann et al. (2025) found 31% incorrect conclusions, Pangakis et al. (2023) demonstrated that automated annotations require task-specific validation, Cheng et al. (2024) propose structured approaches to reduce LLM measurement error, Camuffo et al. (2026) develop variance-aware annotation protocols, Messing (2026) characterizes hidden measurement error in LLM pipelines, and Kim et al. (2025) showed 60% error correlation across models. Audinet de Pieuchon et al. (2025) benchmark debiasing; Keith et al. (2020) survey the broader landscape. These works either restrict to binary labels, cover a single topology, or diagnose quality without connecting it to causal validity.

Sensitivity analysis and correction methods. Our ASV framework draws on the sensitivity analysis without assumptions of Ding and VanderWeele (2016), E-values (VanderWeele and Ding, 2017; Mathur et al., 2018), the robustness value of Cinelli and Hazlett (2020), and the mediation sensitivity analysis of Imai et al. (2010b), and text-based causal inference methods (Pryzant et al., 2021), but parametrizes annotation error severity rather than omitted variable strength or sequential ignorability violations. The quantitative bias analysis framework of Lash et al. (2009) provides a complementary epidemiologic perspective, and Mahajan (2006) establishes identification of regression models with misclassified regressors via instrumental variables. The unified measurement-error and missing-data framework of Blackwell et al. (2017) similarly motivates treating annotation noise as a first-class object of inference. For correction, PPI++ (Angelopoulos et al., 2023) and DSL (Egami et al., 2023) are topology-agnostic; Zrnić and Candès (2024) extend PPI to gold-free settings, Demirel et al. (2024) generalize prediction-powered inference to causal parameters across populations, and Cadei et al. (2025) adapt it to treatment-effect targets via a deconfounded risk objective, and Gligorić et al. (2025) leverage LLM confidence scores. Our contribution reveals that this agnosticism is costly: correction effectiveness is itself topology-dependent, with matrix-based methods provably ineffective for four topologies. Bayesian measurement error models (Richardson and Gilks, 1993; Gustafson, 2004) and EM imputation offer alternatives that bypass FWL invariance.

3 Method

3.1 Setup and Notation

Let A denote a true K-class categorical variable (e.g., sentiment with $K{=}3$ levels) and A^* its observed, potentially misclassified version produced by an LLM annotator. The misclassification process is characterized by a $K{\times}K$ confusion matrix C with entries $C_{jk} = P(A^*{=}j \mid A{=}k)$, where each column sums to one. Following Hernán and Cole (2009), we represent this measurement process using *starred-node* DAGs: the true variable A is latent, and the observed A^* is connected to A via a dashed measurement edge.

Notation conventions. Our analysis characterizes systematic bias—the value an estimator converges to as $N \rightarrow \infty$ (the probability limit, or plim)—rather than finite-sample variance. The Frisch–Waugh–Lovell (FWL) theorem guarantees that each regression coefficient equals the coefficient from a simple regression after residualizing all other variables; this is why matrix-based corrections fail for covariate topologies (Section 4.5): replacing the mismeasured variable with a corrected version does not change the residualized relationship. Table 19 (Appendix C) collects all notation.

Definition 1 (Non-differential misclassification)**.** *Misclassification of A is non-differential with respect to outcome Y and covariates $\mathbf{Z}$ if $P(A^*{=}j \mid A{=}k, Y, \mathbf{Z}) = P(A^*{=}j \mid A{=}k) = C_{jk}$ for all j, k.*

This holds when the LLM classifies without observing the outcome—the typical CSS setup (Ziems et al., 2024); topic-stratified confusion matrices confirm cross-group stability (Appendix B.5). Violations arise when pretraining data encodes A–Y associations; our stress test (Appendix B.6) shows <3 pp degradation under uniform ε=0.05 violations, but ε=0.01 structured departure produces sign reversal in 14/16 $K{\geq}3$ exposure configurations (Appendix B.16).

We study seven canonical DAG topologies (Figure 1) (Pearl, 2009) in which A occupies structurally distinct causal roles: confounder, exposure, mediator, collider, M-bias variable, front-door mediator, and instrument. Throughout, we derive probability limits under *linear* structural equation models (Carroll et al., 2006), the standard framework for exact characterization of topology-dependent bias; robustness under logistic regression is verified in Appendix B.7 and B.14.

3.2 Multi-Class Bias Expressions

For each topology, we derive the plim[2] of the naive estimator that substitutes A^* for A. Let τ denote the target causal parameter—its precise definition is topology-specific (e.g., the coefficient on X for confounding, the category-specific β_j for exposure, or the direct effect τ_{direct} for mediation)—and $\hat{\tau}(A^*)$ the estimator using misclassified annotations.

[2]The probability limit (plim) is the value to which an estimator converges as sample size $N \to \infty$; it characterizes systematic bias separately from sampling variability.

Confounding (A = confounder). When A confounds the $X{\to}Y$ relationship, the OLS estimator controlling for A^* has probability limit

$$\text{plim}\ \hat{\tau}(A^*) = \left(\mathbb{E}[Z'Z]\right)^{-1}\mathbb{E}[Z'X]\,\boldsymbol{\beta}, \qquad (1)$$

where $Z = [X, \text{dummies}(A^*)]$, $\boldsymbol{\beta}$ contains the true coefficients, and the target parameter τ is the coefficient on X. Misclassification produces residual confounding that monotonically approaches the omitted variable bias, but the sign of $\hat{\tau}$ is preserved: $\tau_{\text{sign\_flip}} = 0$ structurally.[3]

Exposure (A = treatment variable). When A is the exposure of interest, the biased coefficient for category j becomes a weighted combination of true coefficients:

$$\text{plim}\ \hat{\beta}_j(A^*) = \sum_{k=1}^{K} w_{jk}\,\beta_k, \qquad (2)$$

where w_{jk} depends on the confusion matrix C and the population marginal $\pi_k = P(A{=}k)$. Unlike the binary case, where non-differential misclassification guarantees attenuation toward the null (Brenner, 1993), the $K{\geq}3$ case permits w_{jk} to assign negative weight to categories with opposite-sign coefficients, producing *sign reversal*. A near-permutation confusion matrix effectively swaps category labels, making sign reversal the dominant outcome (83.4% under uniform Dirichlet prior). This failure mode is structurally impossible at $K{=}2$ because a $2{\times}2$ stochastic matrix with positive diagonal cannot reverse the sign of a single coefficient.

Numerical Verification 2: Primary coefficient preservation. Across 23 empirical LLM configurations, the primary coefficient (largest $|\beta_k|$) never reverses sign, while 61% reverse a secondary coefficient. This reflects magnitude dominance: under 100,000 Dirichlet$(1,1,1)$ matrices with diagonal ≥ 0.7, primary sign-flip rate is 0%; without a diagonal constraint, both coefficients flip ~50% (detailed breakdown in Appendix A.8).

[3]Consistent with Brenner (1993): misclassification degrades confounding control, producing amplification ($|\hat{\tau}| > |\tau|$, universal in this topology) but never sign reversal ($\text{sign}(\hat{\tau}) \neq \text{sign}(\tau)$).

Mediation (A = mediator). When A mediates the $X \to Y$ relationship (VanderWeele, 2015), the direct effect estimator controlling for A^* has probability limit

$$\operatorname{plim} \hat{\tau}_{\text{direct}}(A^*) = \tau_{\text{direct}} + \big(1 - \lambda(C)\big)\, \tau_{\text{indirect}}, \tag{3}$$

where $\lambda(C) \in [0, 1]$ under proportional effects (Appendix A, Proposition 4), with $\lambda(I) = 1$ under perfect classification. Misclassification ($\lambda < 1$) causes the residual indirect effect to be absorbed into the direct path estimate, producing upward bias in $|\hat{\tau}_{\text{direct}}|$ but preserving the sign ($\tau_{\text{sign\_flip}} = 0$).

Collider, M-bias, and front-door. Three topologies exhibit *protective* or *immune* behavior. **Collider:** conditioning on A^* instead of A attenuates the spurious path, so annotation error reduces collider bias ($\tau_{\text{sign\_flip}} = 0$). **M-bias:** in the M-shaped DAG (Greenland, 2003), misclassifying A weakens the M-bias path; this topology is immune (ASV $= \infty$). For instance, annotator expertise (U_1) and topic difficulty (U_2) may independently affect both annotation quality and the treatment-outcome relationship, creating an M-shaped DAG. **Front-door:** the front-door estimator (Bellemare et al., 2024) is attenuated in both $X \to A$ and $A \to Y$ stages, producing moderate bias with $\tau_{\text{sign\_flip}} = 0$.

Instrumental variable (A = instrument). The Wald estimator (Angrist and Imbens, 1995) $\hat{\tau}_{\text{IV}} = \widehat{\text{cov}}(A^*, Y)/\widehat{\text{cov}}(A^*, X)$ has C in the denominator. Misclassification can drive first-stage coefficients toward zero or reverse their sign, causing the ratio estimator to diverge or flip. Under random Dirichlet matrices, 52.6% produce sign reversal, making IV the most fragile topology (detailed analysis including the Dirichlet subspace geometry in Appendix B.4).

Structural sign-flip guarantee.

Proposition 1 (Sign preservation—analytic results)**.** *Under non-differential misclassification with $C_{kk} > 0$ for all k:* (a) *Confounding preserves sign when* $\text{sign}(\tau) = \text{sign}(\tau + B_{\text{OV}})$, *where* $B_{\text{OV}} = \text{Cov}(g(A), f(A))/\text{Var}(T)$ *is the omitted-variable bias (Cauchy–Schwarz; Appendix A.3).* (b) *Collider preserves sign when* $|\tau| > \sqrt{\text{Var}(g)\,\text{Var}(h)}/\mathbb{E}[\text{Var}(T|A)]$ *(Berkson attenuation; Appendix A.6).* (c) *Exposure can produce sign reversal for $K \geq 3$, even with $C_{kk} > 0.5$ (counterexample in Appendix A.8).*

Our theoretical results establish a three-tier hierarchy: (i) analytically proved (Proposition 1(a,b), Theorem 1), (ii) numerically verified with zero violations across 20,000+ random configurations (Observation 1, Numerical Verifications 1–2), and (iii) conjectured with ε-approximate bounds (Conjecture 1).

Numerical Verification 1: Non-exposure sign preservation. Sign preservation holds under topology-specific conditions (mediation: $\text{sign}(\tau_{\text{dir}}) = \text{sign}(\tau_{\text{total}})$; front-door: $\text{sign}(\lambda\alpha_U) = \text{sign}(\lambda\alpha_U + \text{Cov}(T, f(A)))$; M-bias: attenuating for all C). Verified across 100,000 random matrices with zero violations (Table 2, Appendix A.9); analytic proofs for general K remain open. Since these conditions involve true DGP parameters, the ASV framework (§3.3) provides a complementary assessment.

3.3 Annotation Sensitivity Values

How much annotation error can a study tolerate before conclusions change? We adapt the sensitivity analysis paradigm—E-values (VanderWeele and Ding, 2017) (the minimum confounding strength needed to explain away an observed effect), partial-R^2 bounds (Cinelli and Hazlett, 2020)—to $K \times K$ annotation error, defining Annotation Sensitivity Values (ASVs). Our approach: (1) prove ε-approximate monotonicity along a 1D path $C(\delta)$ (Theorem 1), reducing the K^2-dimensional problem to a scalar threshold; (2) define three ASV diagnostics addressing topology-dependent failure modes; (3) combine ASVs with FWL invariance to route practitioners to topology-appropriate corrections.

Parameterization. We parameterize the confusion matrix as $C(\delta) = (1-\delta)I + \delta C_0$, where C_0 is a reference confusion matrix and $\delta \in [0, 1]$ controls severity (δ=0: perfect; δ=1: matches C_0). Non-negativity requires $\delta \leq \delta_{\max} = \min_j 1/(1 - (C_0)_{jj}) \geq 1$; we recommend bootstrapping C_0 and reporting ASV confidence intervals (Appendix B.8).

Three ASV types. We define three complementary measures: (1) **Magnitude ASV**

$\delta^*_{\text{mag}} = \min\{\delta : |\text{plim}\ \hat{\tau}(C(\delta)) - \tau| > \epsilon\}$—analogous to E-values; (2) **Significance ASV** $\delta^*_{\text{sig}} = \min\{\delta : |\text{plim}\ \hat{\tau}(C(\delta))| < z_{\alpha/2} \cdot SE(N)\}$—relevant for attenuating topologies; (3) **Sign-flip ASV** $\delta^*_{\text{sf}} = \min\{\delta : \text{sign}(\hat{\tau}) \neq \text{sign}(\tau)\}$—exposure and IV only. The significance ASV depends on N through $SE(N)$, exhibiting a sharp phase transition at a critical N^* below which all rejections are blocked.

Theorem 1 (Envelope monotonicity—K=2)**.** *For K=2 confounding and mediation topologies with proportional effects ($f = \lambda g$ for some scalar λ) under $C(\delta) = (1-\delta)I + \delta C_0$ with $(C_0)_{jj} \geq \frac{1}{2}$, $|B(C(\delta))|$ is monotonically non-decreasing in $\delta \in [0, 1]$.*

The proof (Appendix A.4, A.5) exploits perspective function convexity; $(C_0)_{jj} \geq \frac{1}{2}$ excludes degenerate matrices. Proportional effects ($f = \lambda g$) holds when a single latent factor drives both paths—a natural assumption in CSS annotation tasks, where a single quality dimension typically governs both the treatment–covariate and mediator–outcome relationships. Under NLP-realistic DGP conditions, the out-of-bound rate drops to 0.07% with zero sign failures (Table 12); even under non-proportional effects with 23 real LLM matrices, sign preservation remains ≥97.3% (Table 16). Failures concentrate in extreme asymmetric DGPs (Sparse: 24.2% OOB) unlikely in balanced NLP classification. When $\lambda(C) \notin [0, 1]$, Proposition 3 provides a complementary bound ($|\tau| > \bar{B}$).

Observation 1 (Envelope monotonicity—K≥3)**.** *For K≥3 under the same conditions, numerical verification across 20,000 random (C_0, p, g) triples per K shows worst-case deviations are negligible (≤ 0.013; Table 10 in Appendix B.12).*

Conjecture 1 (ε-approximate envelope monotonicity)**.** *For non-IV topologies under $C(\delta) = (1-\delta)I + \delta C_0$, $|B(C(\delta))|$ is monotonic up to $\varepsilon < 0.06$ deviations under default DGP parameters (collider/M-bias non-increasing, others non-decreasing; see Appendix B.12 for DGP-dependent bounds). Verified across 10,000 random matrices per topology for each $K \in \{3, 5, 7\}$ (210,000 total), $\tau \in \{0.1, 0.5, 1.0, 2.0, 5.0\}$.*

Remark 1 (Structural distinctions from E-values)**.** *ASV diagnostics differ from E-values (VanderWeele and Ding, 2017) in three respects (Table 18, Appendix):* (i) Topology dependence*: the same C preserves sign under confounding but can reverse it under exposure (Table 1), whereas E-values treat all confounding identically.* (ii) Sign-flip diagnostic*: E-values diagnose magnitude robustness but not sign reversal; the sign-flip ASV δ^*_{sf} (Algorithm 1) has no E-value counterpart.* (iii) Parametric structure*: ASV reduces K^2-dimensional confusion matrix space to one dimension via $C(\delta)$, but $B(\delta)$ is topology-specific and can be non-monotone (IV).*

For K=2, Theorem 1 guarantees exact monotonicity; for K≥3, monotonicity rests on numerical evidence (Conjecture 1), with ASV thresholds robust to ε-approximate deviations ($\varepsilon < 0.06$ under default DGP; Appendix B.12). Sign-flip ASV < 1 flags sign-reversal-prone pairs; magnitude ASV $< \delta_{\text{observed}}$ indicates bias exceeds tolerance (Algorithm 1). Rankings are stable across DGP parameterizations (Appendix B.11); binarization introduces additional topology-dependent bias (Appendix B.1).

4 Experiments

We validate through Monte Carlo simulation (K=3), cross-K robustness checks (K=5, 7), and empirical validation with real LLM confusion matrices from three NLP tasks (K=2, 3, 10).

4.1 Setup

For each of the seven topologies (confounding, exposure, mediation, collider, M-bias, front-door, IV), we sample 100,000 confusion matrices from a Dirichlet(1, 1, 1) prior, compute the analytical probability limit of the causal estimator, and validate against Monte Carlo simulation (N=10,000). ASV experiments use $C(\delta) = (1-\delta)I + \delta C_0$.

4.2 Bias Taxonomy

Theory-MC agreement. Analytical probability limits match Monte Carlo estimates with grand MAPE of 1.6% across 161 scenarios (23 LLM configurations × 7 topologies, N=10,000). Per-topology MAPE ranges from 0.3% (confounding) to 3.7% (IV); the IV maximum reflects the Wald estimator's small-denominator variance amplification.

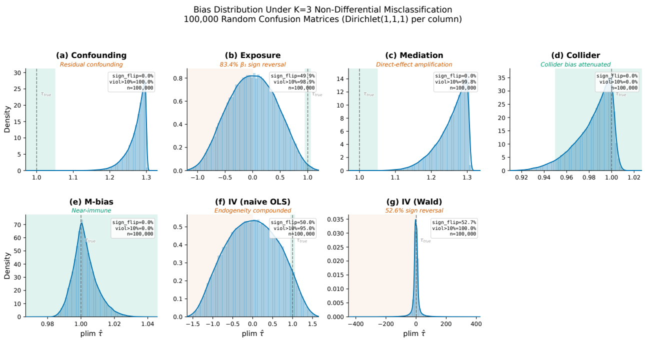

Figure 2: Bias distributions under 100,000 random K=3 confusion matrices. Red = sign reversal. Exposure: 83.4% sign reversal; IV: 52.6%.

Exposure DAG: sign reversal under multi-class misclassification. Under realistic LLM accuracy (diagonal ≥ 0.7), the violation rate is 13.7% with 2.1% sign reversal; even at diagonal ≥ 0.9, violation persists at 11.9%. Under the full Dirichlet$(1, 1, 1)$ prior, 85% of K=3 confusion matrices violate the binary "toward-null" guarantee, and 83.4% cause sign reversal of at least one coefficient. The K=2 control confirms that all tested binary matrices maintain classical attenuation (Brenner, 1993), establishing that the multi-class failure is structurally distinct.

Structural guarantee for non-exposure topologies. Five topologies exhibit zero sign-flip rate (Table 1): confounding, mediation, and front-door show 100% amplification (Brenner, 1993; Dosemeci et al., 1990) but never reverse the sign; collider is protective ($|B(C)|$ non-increasing; Conjecture 1); M-bias shows weakest sensitivity (53.2% violation). All patterns persist at K=5, 7 (Appendix B.9).

IV: Wald estimator instability. The IV topology is the sole non-exposure exception: 52.6% of random matrices cause sign reversal, stable across K=3, 5, 7. Under the full Dirichlet$(1, 1, 1)$ prior, diagonal dominance ($C_{kk} > 0.8$) reduces but does not eliminate IV sign reversal (Appendix B.4); however, across all 23 empirical LLM configurations—including low-accuracy annotators (mean diagonal as low as 0.45)—IV exhibits zero sign reversal. For IV designs where annotation error is unavoidable, practitioners may consider bounds approaches (Manski, 1990), robust IV methods, or two-sample designs where the instrument is measured without error in a separate sample.

The bias distributions are visualized in Figure 2.

Table 1: Bias under K=3 non-differential misclassification (100,000 Dirichlet$(1, 1, 1)$ matrices). τ_{sf}=0?: sign-flip guarantee holds unconditionally (Yes), conditionally on proportional effects (Cond.; Proposition 1), or not (No). Amplif. denotes $|\hat{\tau}(A^*)| > |\tau|$. For collider, 100% amplification reflects Berkson bias from conditioning, not error amplification—misclassification is protective (Section 3). Violation: $|B| > 0$; Sign Flip: $\text{sign}(\hat{\tau}) \neq \text{sign}(\tau)$.

Topology	Violation	Sign Flip	Amplif.	τ_{sf}=0?
Confounding	100%	0%	100%	Cond.
Exposure	85%	83.4%	14.1%	No
Mediation	100%	0%	100%	Cond.
Front-door	100%	0%	100%	Cond.
Collider	100%	0%	100%	Cond.
M-bias	53.2%	0%	10.4%	Yes
IV	100%	52.6%	100%	No

4.3 ASV Framework Validation

Envelope monotonicity and magnitude ASV. The ASV envelope is ε-approximately monotonic for non-IV topologies ($\varepsilon < 0.06$ under default DGP; Conjecture 1), confirming δ as a meaningful severity dial. IV is excluded due to first-stage singularities. Table 3 reports magnitude ASV ranges at τ=0.5 with a 10% bias threshold, ranking from most fragile to most robust: front-door (0.064–0.387) < mediation < confounding < IV (0.101–0.607) < collider (0.338–2.028) < M-bias (∞). M-bias achieves infinite ASV because annotation error only reduces the already-small spurious path.

Significance and sign-flip ASV. At τ=0.05, significance ASV achieves 100% power for confounding, front-door, and M-bias, but 0% for mediation, collider, and IV (correct behavior; Type I error is 0%). Sign-flip ASV: 83.4% of exposure matrices yield $\tau_{\text{sign\_flip}} < 1$;

IV ranges 1.1–6.7. The complete ranking (Figure 4, Table 3, Appendix B) confirms IV as maximally fragile and M-bias as immune (ASV $= \infty$).

N-sensitivity. Significance ASV exhibits a sharp phase transition at $N^* \approx$ 6,000–10,000, below which all rejections are blocked. Bootstrap CIs (B=1,000; Appendix B.8): $\delta^* = 0.94\,[0.60, 1.36]$ (exposure), $0.75\,[0.44, 1.29]$ (IV); despite wide IV intervals from denominator instability, the topology ranking is stable across all replicates.

4.4 Empirical Validation

We validate using real LLM confusion matrices from HumAID (Alam et al., 2021) (K=10), VAST (K=3), and CivilComments (K=2), with four LLMs producing 23 configurations under zero-shot and few-shot prompting (161 bias scenarios).

Empirical bias ranking. HumAID (K=10) matches predictions: IV (12.4–42.1%) and exposure (9.0–17.4%) dominate; remaining topologies <15%. Bias varies substantially across K: VAST (K=3) IV reaches 326% versus 42% for HumAID. Across all 56 HumAID scenarios (K=10), F_1 and |bias| show no significant rank correlation (Spearman $\rho = -0.10$, 95% CI $[-0.36, 0.17]$, $p = 0.45$; power to detect $|\rho| \geq 0.3$: 0.57; Figure 3).

Of all $K-1$ exposure coefficients, 14/23 configurations (61%) exhibit sign reversal—all in secondary coefficients (Numerical Verification 2); zero across 138 non-exposure scenarios. ASV diagnostics flag 55.4% of HumAID and 100% of CivilComments as magnitude-unsafe (Appendix B.10).

4.5 Correction Guidance

We evaluate five correction methods (naive plug-in, MLA, DSL (Egami et al., 2023), MC-SIMEX, PPI++ (Angelopoulos et al., 2023)) across all seven topologies under six error structures (210 configurations; full results in Table 4, Appendix B).

FWL invariance. Under linear OLS, FWL implies that any linear transformation of $\mathbf{D}^*$ preserves the column space relative to T for the four covariate topologies (where A is a con-

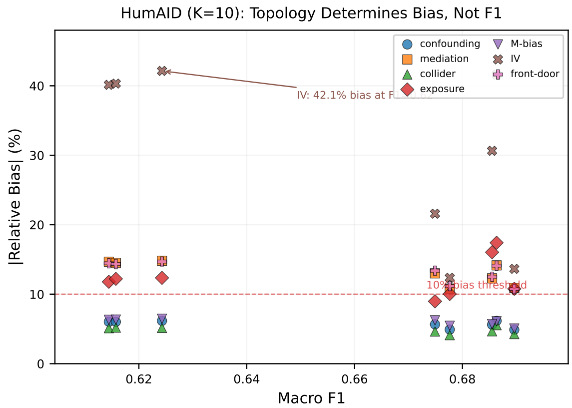

Figure 3: Topology determines downstream causal bias, not F_1. HumAID K=10 (56 points): within F_1 range 0.614–0.690, bias varies 10× (IV 42.1% vs. collider <5.6%).

trol, not the treatment), so naive plug-in and MLA achieve exactly 0.0% bias reduction (Appendix B.2); exposure and IV are excluded. Under logistic regression the invariance breaks but empirically by < 0.05% (Table 13); all structural patterns persist ($|\Delta\,\mathrm{SF}_\tau| < 0.02$; Appendix B.7).

Method ranking. At K=3, PPI++ achieves the highest grand-mean RMSE reduction (47.0%) but requires gold labels and is harmful for M-bias (−88%): misclassification attenuates the spurious M-bias path, and PPI++ removes this protective attenuation. PPI++ also degrades with K, becoming harmful for exposure at K=10. MC-SIMEX is more K-robust (27–70% reduction across K=3, 5, 10) and gold-free, though it harms front-door (−62.4%). Rankings are stable across K (Spearman $\rho \geq 0.80$; Appendix B.19); no single method dominates.

Case studies and sensitivity. ClinicalBERT validation (Appendix B.3) confirms FWL invariance in clinical NLP. PPI++ requires ≥20% gold labels for IV but harms M-bias (−125% RMSE at 5% gold; Appendix B.15). ASV case studies show 3× higher error tolerance for confounder vs. exposure (Appendix B.18).

5 Conclusion

We characterized how multi-class annotation errors interact with causal DAG topology: the binary attenuation guarantee breaks at $K \geq 3$, where exposure and IV can reverse signs while five topologies preserve them. ASVs yield a vulnerability ranking from IV (most fragile) to M-bias (immune); matrix-based corrections are provably ineffective for four covariate topologies due to FWL invariance. Code and data will be released.

Limitations

Our framework assumes non-differential misclassification ($P(A^*|A, Y, \mathbf{Z}) = P(A^*|A)$), which holds when the LLM does not observe Y but may be violated when it accesses Y-correlated information; a stress test (Appendix B.6) shows that structured violation patterns can exceed or fall below uniform predictions. Closed-form expressions are derived under $K=3$ linear structural equations; extensions to arbitrary K and nonlinear models (Schennach, 2020) remain open (Appendix B.7), though correction method rankings are stable across $K=3, 5, 10$ (Spearman $\rho \geq 0.80$, with PPI++ degrading at $K=10$; Appendix B.19). The ASV diagonal parameterization $C(\delta) = (1-\delta)I + \delta C_0$ assumes uniform degradation across categories; rare-class-specific degradation would require multi-dimensional extensions. Algorithm 1 thresholds are calibrated under our default DGP; practitioners should re-calibrate under domain-specific structural parameters. When the true DAG is uncertain, practitioners can apply Algorithm 1 under each candidate topology and report the range of conclusions as a form of specification sensitivity analysis. FWL invariance holds under linear OLS and approximately under logistic regression; applicability to non-parametric methods (doubly robust estimators, TMLE) remains unverified. Our framework assumes C is known or estimable from held-out gold-standard labels; obtaining accurate C itself requires gold annotations, a bootstrapping constraint mitigated by cross-validation or expert subset labeling. Our 23 configurations span four open-weight LLMs (7B–235B) with diagonal accuracy 0.45–0.81; commercial models and ordinal tasks remain untested.

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A Proofs of Proposition 1 and Theorem 1

We provide algebraic proofs of sign preservation for the confounding and collider topologies (Proposition 1(a)–(b)), an explicit counterexample for exposure (Proposition 1(c)), numerical verification for mediation, M-bias, and front-door (Numerical Verification 1), and the monotonicity proof for confounding (Theorem 1). Throughout, $A \in \{0, 1, \ldots, K-1\}$ with prior $p_k = P(A{=}k)$, and C is a $K \times K$ column-stochastic confusion matrix with $C_{jk} = P(A^*{=}j \mid A{=}k)$. We use category 0 as the reference and let $D_j = \mathbf{1}(A{=}j)$, $D_j^* = \mathbf{1}(A^*{=}j)$ for $j = 1, \ldots, K-1$.

A.1 Notation and Key Identities

Under non-differential misclassification, the misclassified moments factor linearly through C. Define the population vectors:

$$q_j = P(A^*{=}j) = \textstyle\sum_k C_{jk}\, p_k, \tag{4}$$
$$\mathrm{ct}_j = \mathbb{E}[T \cdot D_j^*] = \textstyle\sum_k C_{jk}\, \mathbb{E}[T \cdot D_k], \tag{5}$$
$$\mathrm{cy}_j = \mathbb{E}[Y \cdot D_j^*] = \textstyle\sum_k C_{jk}\, \mathbb{E}[Y \cdot D_k]. \tag{6}$$

Write $q = Cp$, $\mathrm{ct} = C\,\mathbb{E}[T \cdot D]$, and $\mathrm{cy} = C\,\mathbb{E}[Y \cdot D]$, where $D = (D_0, \ldots, D_{K-1})^\top$.

A.2 Frisch–Waugh–Lovell Decomposition

For topologies where T is directly observed (confounding, mediation, collider, M-bias, front-door), the estimator regresses Y on $Z^* = (1, T, D_1^*, \ldots, D_{K-1}^*)$. By the Frisch–Waugh–Lovell (FWL) theorem, the coefficient on T satisfies:

$$\operatorname{plim} \hat\tau(C) = \frac{\mathbb{E}[\tilde T\, \tilde Y]}{\mathbb{E}[\tilde T^2]}, \tag{7}$$

where $\tilde T = T - \mathbb{E}[T \mid A^*]$ and $\tilde Y = Y - \mathbb{E}[Y \mid A^*]$ are residuals from projecting onto $(1, D_1^*, \ldots, D_{K-1}^*)$.

Lemma 1 (Bias decomposition)**.** *Suppose $Y = \tau T + f(A) + \varepsilon_Y$ with $\varepsilon_Y \perp (T, A, A^*)$. Then*

$$\operatorname{plim} \hat\tau(C) = \tau + \frac{R(C)}{D(C)}, \tag{8}$$

where

$$R(C) = \mathbb{E}\big[\mathrm{Cov}(T,\, f(A) \mid A^*)\big], \tag{9}$$
$$D(C) = \mathbb{E}\big[\mathrm{Var}(T \mid A^*)\big] > 0. \tag{10}$$

Proof. Expand $\tilde Y$ using $Y = \tau T + f(A) + \varepsilon_Y$:

$$\tilde Y = \tau\, \tilde T + \big(f(A) - \mathbb{E}[f(A) \mid A^*]\big) + \varepsilon_Y.$$

Since $\tilde T$ is measurable w.r.t. (T, A^*) and $\varepsilon_Y \perp (T, A, A^*)$:

$$\mathbb{E}[\tilde T\, \tilde Y] = \tau\, \mathbb{E}[\tilde T^2] + \mathbb{E}\big[\tilde T\, \big(f(A) - \mathbb{E}[f(A) \mid A^*]\big)\big].$$

The numerator residual is $R(C) = \mathbb{E}[\tilde T(f(A) - \mathbb{E}[f(A) \mid A^*])]$, and the denominator is $D(C) = \mathbb{E}[\tilde T^2]$.

By the law of total covariance:

$$R(C) = \sum_{j=0}^{K-1} q_j\ \mathrm{Cov}(T,\, f(A) \mid A^*{=}j) = \mathbb{E}[\mathrm{Cov}(T,\, f(A) \mid A^*)].$$

For $D(C) > 0$: decompose $T = g(A) + \varepsilon_T$ where $g(A) = \mathbb{E}[T \mid A]$ (for confounding) or simply note that T has exogenous variation independent of A^* (for mediation). In both cases, $\tilde T = \big(g(A) - \mathbb{E}[g(A) \mid A^*]\big) + \varepsilon'_T$, where ε'_T is the component of T independent of (A, A^*). Hence $D(C) \geq \mathrm{Var}(\varepsilon'_T) > 0$. □

A.3 Confounding Topology (A = Confounder)

Data-generating process. $A \to T$, $A \to Y$, $T \to Y$. Using dummy coding with category 0 as reference:

$$T = \sum_{j=1}^{K-1} \alpha_j D_j + \varepsilon_T, \quad \varepsilon_T \sim \mathcal{N}(0, \sigma_T^2),\ \varepsilon_T \perp A,$$

$$Y = \tau\, T + \sum_{j=1}^{K-1} \beta_j^{(A)} D_j + \varepsilon_Y, \quad \varepsilon_Y \sim \mathcal{N}(0, 1),\ \varepsilon_Y \perp (T, A).$$

Define $g(k) = \sum_j \alpha_j \mathbf{1}(k{=}j)$ and $f(k) = \sum_j \beta_j^{(A)} \mathbf{1}(k{=}j)$, so $g(0) = f(0) = 0$.

Closed-form plim. Using the FWL decomposition, the coefficient on T is:

$$\operatorname{plim} \hat\tau(C) = \frac{\mu_{TY} - \sum_{j=0}^{K-1} \frac{\mathrm{ct}_j\, \mathrm{cy}_j}{q_j}}{s_T - \sum_{j=0}^{K-1} \frac{\mathrm{ct}_j^2}{q_j}}, \tag{11}$$

where $s_T = \mathbb{E}[T^2]$, $\mu_{TY} = \mathbb{E}[TY]$, and ct_j, cy_j, q_j are defined in (4)–(6).

Derivation. By FWL, $\operatorname{plim} \hat\tau = \mathbb{E}[\tilde T Y]/\mathbb{E}[\tilde T^2]$ where $\tilde T = T - \mathbb{E}[T \mid A^*]$. Since $\mathbb{E}[T \mid A^*{=}j] = \mathrm{ct}_j/q_j$:

$$\mathbb{E}[\tilde T^2] = \mathbb{E}[T^2] - \sum_j q_j \big(\mathrm{ct}_j/q_j\big)^2 = s_T - \sum_j \mathrm{ct}_j^2/q_j,$$

$$\mathbb{E}[\tilde T\, Y] = \mathbb{E}[TY] - \sum_j \big(\mathrm{ct}_j/q_j\big)\, \mathbb{E}[Y \cdot D_j^*] = \mu_{TY} - \sum_j \mathrm{ct}_j\, \mathrm{cy}_j/$$

□

Verification at $C = I$. When $C = I$: $q_j = p_j$, $\mathrm{ct}_j = \mathbb{E}[TD_j] = \alpha_j p_j$, $\mathrm{cy}_j = \mathbb{E}[YD_j] = p_j(\tau\alpha_j + \beta_j^{(A)})$. Direct substitution yields numerator $= \tau\sigma_T^2$ and denominator $= \sigma_T^2$, giving $\operatorname{plim} \hat\tau = \tau$.

Bias structure. By Lemma 1:

$$\operatorname{plim} \hat\tau(C) = \tau + B(C), \quad B(C) = \frac{\mathbb{E}[\mathrm{Cov}(g(A),\, f(A) \mid A^*)]}{D(C)}. \tag{12}$$

By the law of total covariance:

$$\mathbb{E}[\mathrm{Cov}(g, f \mid A^*)] = \mathrm{Cov}(g(A), f(A)) - \mathrm{Cov}\big(\mathbb{E}[g(A) \mid A^*], \mathbb{E}[f(A) \mid A^*]\big) \tag{13}$$

At $C = I$: $\mathbb{E}[g \mid A^*] = g(A)$, so the between-group covariance equals $\mathrm{Cov}(g, f)$ and $B(I) = 0$.

At $C = \frac{1}{K}\mathbf{1}\mathbf{1}^\top$ (uniform, $A^* \perp A$): $\mathbb{E}[g \mid A^*]$ is constant, so the between-group covariance is zero and

$$B_{\mathrm{OV}} \;\equiv\; B\big(\tfrac{1}{K}\mathbf{1}\mathbf{1}^\top\big) = \frac{\mathrm{Cov}(g(A),\, f(A))}{\mathrm{Var}(T)}, \tag{14}$$

which equals the classical omitted-variable bias (no control for A).

Sign preservation condition. Since $D(C) > 0$, $\mathrm{sign}(\mathrm{plim}\,\hat{\tau}) = \mathrm{sign}(\tau \cdot D(C) + R(C))$. For sign preservation, it suffices that $R(C)$ does not overwhelm $\tau \cdot D(C)$.

Proposition 2 (Confounding sign preservation — proportional case)**.** *If $f(A) = \lambda\, g(A)$ for some constant λ (i.e., the confounder effects on T and Y are proportional), then*

$$B(C) = \big(1 - \xi(C)\big)\, B_{\mathrm{OV}}, \tag{15}$$

where

$$\xi(C) = \frac{\mathrm{Var}(\mathbb{E}[g(A) \mid A^*])\; \sigma_T^2}{\mathrm{Var}(g(A))\big(\mathrm{Var}(g(A)) - \mathrm{Var}(\mathbb{E}[g(A) \mid A^*]) + \sigma_T^2\big)} \in [0, 1].$$

Consequently, $B(C)$ lies between 0 and B_{OV}, and $\mathrm{sign}(\mathrm{plim}\,\hat{\tau}) = \mathrm{sign}(\tau)$ whenever $\mathrm{sign}(\tau) = \mathrm{sign}(\tau + B_{\mathrm{OV}})$.

Proof. When $f = \lambda g$: the between-group covariance in (13) becomes $\lambda\, \mathrm{Var}(\mathbb{E}[g \mid A^*])$, and $\mathrm{Cov}(g, f) = \lambda\, \mathrm{Var}(g)$. So $R(C) = \lambda\big(\mathrm{Var}(g) - \mathrm{Var}(\mathbb{E}[g \mid A^*])\big) = \lambda\, V_W$, where $V_W = \mathrm{Var}(g) - V_B$ with $V_B = \mathrm{Var}(\mathbb{E}[g \mid A^*])$. Similarly, $D(C) = V_W + \sigma_T^2$. Hence

$$B(C) = \frac{\lambda\, V_W}{V_W + \sigma_T^2}, \qquad B_{\mathrm{OV}} = \frac{\lambda\, V}{V + \sigma_T^2},$$

where $V = \mathrm{Var}(g) = V_B + V_W$. The ratio

$$\frac{B(C)}{B_{\mathrm{OV}}} = \frac{V_W(V + \sigma_T^2)}{V(V_W + \sigma_T^2)} = 1 - \xi(C)$$

is a decreasing function of $V_B \in [0, V]$. At $V_B = 0$ (no classification, C uniform): $\xi = 0$, $B = B_{\mathrm{OV}}$. At $V_B = V$ (perfect classification, $C = I$): $\xi = 1$, $B = 0$. Since $0 \le V_B \le V$, we have $\xi \in [0, 1]$ and $B(C) \in [0, B_{\mathrm{OV}}]$ (with the sign of B_{OV}). $\square$

Proposition 3 (Confounding sign preservation — general case)**.** *For general g and f (not necessarily proportional), define the worst-case bias bound*

$$\bar{B} \;=\; \frac{\sqrt{\mathrm{Var}(g(A))\; \mathrm{Var}(f(A))}}{\sigma_T^2}. \tag{16}$$

If $|\tau| > \bar{B}$, then $\mathrm{sign}(\mathrm{plim}\,\hat{\tau}(C)) = \mathrm{sign}(\tau)$ for all column-stochastic C.

Proof. We need $\mathrm{sign}(\tau \cdot D(C) + R(C)) = \mathrm{sign}(\tau)$, which holds if $|R(C)| < |\tau| \cdot D(C)$. Since $D(C) \ge \sigma_T^2$ (shown in Lemma 1), it suffices to show $|R(C)| < |\tau|\, \sigma_T^2$.

Step 1: Bound $|R(C)|$. $R(C) = \mathbb{E}[\mathrm{Cov}(g(A), f(A) \mid A^*)]$. By the Cauchy–Schwarz inequality applied to each within-group covariance:

$$|\,\mathrm{Cov}(g, f \mid A^*{=}j)| \le \sqrt{\mathrm{Var}(g \mid A^*{=}j)\; \mathrm{Var}(f \mid A^*{=}j)}.$$

Taking expectations and applying Cauchy–Schwarz again:

$$\begin{aligned} |R(C)| &\le \mathbb{E}\big[\sqrt{\mathrm{Var}(g \mid A^*)\; \mathrm{Var}(f \mid A^*)}\big] \\ &\le \sqrt{\mathbb{E}[\mathrm{Var}(g \mid A^*)]\; \mathbb{E}[\mathrm{Var}(f \mid A^*)]} \\ &\le \sqrt{\mathrm{Var}(g(A))\; \mathrm{Var}(f(A))}, \end{aligned}$$

where the last step uses $\mathbb{E}[\mathrm{Var}(X \mid Y)] \le \mathrm{Var}(X)$ (law of total variance).

Step 2: Conclude. The bias satisfies

$$|B(C)| = \frac{|R(C)|}{D(C)} \le \frac{\sqrt{\mathrm{Var}(g)\; \mathrm{Var}(f)}}{\sigma_T^2} = \bar{B}.$$

When $|\tau| > \bar{B}$, the bias cannot cancel τ, so $\mathrm{sign}(\mathrm{plim}\,\hat{\tau}) = \mathrm{sign}(\tau)$ for all C. $\square$

Relationship to the omitted-variable condition. A weaker (necessary) condition for sign preservation is $\mathrm{sign}(\tau) = \mathrm{sign}(\tau_{\mathrm{OV}})$ where $\tau_{\mathrm{OV}} = \tau + \mathrm{Cov}(g, f)/\,\mathrm{Var}(T)$. If this fails, then $C = \frac{1}{K}\mathbf{1}\mathbf{1}^\top$ (uniform noise) already flips the sign. The rigorous sufficient condition $|\tau| > \bar{B}$ implies the weaker condition, since $|B_{\mathrm{OV}}| = |\,\mathrm{Cov}(g, f)|/\,\mathrm{Var}(T) \le \bar{B}$ by Cauchy–Schwarz and $\mathrm{Var}(T) \ge \sigma_T^2$.

Verification for paper parameters (K=3). With τ=2.0, $\boldsymbol{\alpha}$=(1.5, −1.0), $\boldsymbol{\beta}^{(A)}$=(1.0, −0.5), σ_T=1.0, $\mathbf{p}$=(0.40, 0.35, 0.25):

$$\mathrm{Var}(g) = \sum_k p_k \alpha_k^2 - \big(\textstyle\sum_k p_k \alpha_k\big)^2 = 0.962,$$
$$\mathrm{Var}(f) = \sum_k p_k (\beta_k^{(A)})^2 - \big(\textstyle\sum_k p_k \beta_k^{(A)}\big)^2 = 0.362,$$
$$\bar{B} = \sqrt{0.962 \times 0.362}\ /\ 1.0^2 = 0.590.$$

Since $|\tau|$=2.0 $>$ $\bar{B}$=0.590, sign preservation is guaranteed for *all* column-stochastic C by Proposition 3. Additionally, $B_{\mathrm{OV}} = \mathrm{Cov}(g, f)/\mathrm{Var}(T) = 0.588/1.962 = +0.300$, giving τ_{OV}=2.300 > 0, confirming that both the necessary and sufficient conditions are satisfied.

Extension to $K > 3$. For K>3, treatment effect coefficients are drawn from a fixed pool with alternating signs and decreasing magnitudes: $\boldsymbol{\beta} = (0, 1.0, -0.5, 0.7, -0.3, 0.5, -0.2, 0.4, -0.15, 0.3, \ldots)$, truncated to K entries. The category distribution uses $p_k \propto 1/(k{+}1)$, normalized to sum to 1. Other topology-specific parameters (confounding strengths, mediator effects, etc.) follow analogous alternating-sign pools. For K=10, $\boldsymbol{\beta} = (0, 1.0, -0.5, 0.7, -0.3, 0.5, -0.2, 0.4, -0.15, 0.3)$.

A.4 Proof of Theorem 1 (Confounding Monotonicity)

Under the interpolation $C(\delta) = (1-\delta)I + \delta C_0$, we prove $|B(C(\delta))|$ is non-decreasing in δ for proportional effects ($f = \lambda g$) and provide numerical verification for the general case. The proof proceeds by direct algebraic analysis of $V_B(\delta) = \mathrm{Var}(\mathbb{E}[g(A) \mid A^*_\delta])$ as a rational function of δ, establishing convexity and bounding the derivative at both endpoints.

Proportional case ($f = \lambda g$). From Proposition 2, $B(C) = \lambda V_W/(V_W + \sigma_T^2)$ where $V_W = \mathrm{Var}(g) - V_B$ and $V_B(\delta) = \mathrm{Var}(\mathbb{E}[g(A) \mid A^*_\delta])$.

Step 1: Convexity of $V_B(\delta)$. Define $N_j(\delta) = \sum_k g_k\, C(\delta)_{jk}\, p_k$ and $q_j(\delta) = \sum_k C(\delta)_{jk}\, p_k$. Both are affine in δ, with $q_j(\delta) > 0$. Then $V_B(\delta) = \sum_j N_j(\delta)^2/q_j(\delta) - \bar{g}^2$. Each term N_j^2/q_j is the composition of the perspective function $(x, t) \mapsto x^2/t$ (convex for $t > 0$) with the affine map $\delta \mapsto (N_j(\delta), q_j(\delta))$, hence convex in δ. A sum of convex functions is convex, so $V_B(\delta)$ is convex on $[0, \delta_{\max}]$.

Step 2: Derivative formula. Let $\mu_j(\delta) = N_j(\delta)/q_j(\delta) = \mathbb{E}[g(A) \mid A^*{=}j]$. Then

$$\frac{dV_B}{d\delta} = -\sum_{j,k} \big[(C_0)_{jk} - \mathbf{1}_{j=k}\big]\, p_k\, (g_k - \mu_j(\delta))^2. \tag{17}$$

Derivation. $\frac{d}{d\delta}\frac{N_j^2}{q_j} = 2\mu_j \dot{N}_j - \mu_j^2 \dot{q}_j$ where $\dot{N}_j = m_j^0 - g_j p_j$ and $\dot{q}_j = q_j^0 - p_j$ (with $m_j^0 = \sum_k g_k (C_0)_{jk} p_k$, $q_j^0 = \sum_k (C_0)_{jk} p_k$). Substituting and using column-stochasticity ($\sum_j (C_0)_{jk} = \sum_j \mathbf{1}_{j=k} = 1$) to cancel the g_k^2 terms yields (17).

Step 3: $V_B'(0) \le 0$ (universal, no conditions on C_0). At δ=0: $\mu_j(0) = g_j$ (since $C(0) = I$), so $(g_k - \mu_j(0))^2 = (g_k - g_j)^2$. The diagonal terms ($j$=$k$) vanish. Hence

$$V_B'(0) = -\sum_{j \ne k} (C_0)_{jk}\, p_k\, (g_j - g_k)^2 \ \le\ 0,$$

since every term is non-negative.

Step 4: $V_B'(1) \le 0$ for K=2. For K=2 with $C_0 = \left[\begin{smallmatrix} a & 1-b \\ 1-a & b \end{smallmatrix}\right]$ and $a, b \ge \frac{1}{2}$, set g_0=0, g_1=1 without loss of generality (the general case scales by $(\Delta g)^2$). At δ=1: q_0=ap_0+$(1-b)p_1$, q_1=$(1-a)p_0$+bp_1, μ_0=$(1-b)p_1/q_0$, μ_1=bp_1/q_1. Expanding (17) at δ=1 and factoring:

$$\begin{aligned} V_B'(1) &= (1{-}a)\, p_0(\mu_0^2{-}\mu_1^2) + (1{-}b)\, p_1\big[(1{-}\mu_1)^2{-}(1{-}\mu_0)^2\big] \\ &= (\mu_0{-}\mu_1)\, \big[(1{-}a)\, p_0(\mu_0{+}\mu_1) + (1{-}b)\, p_1(2{-}\mu_0{-}\mu_1)\big]. \end{aligned}$$

The bracket is non-negative (since $0 \le \mu_j \le 1$ and all coefficients are non-negative). Direct calculation gives $\mu_0{-}\mu_1 = p_0\, p_1(1{-}a{-}b)/(q_0\, q_1)$ (expand $(1{-}b)\, q_1 - b\, q_0 = p_0(1{-}a{-}b)$). Under $a, b \ge \frac{1}{2}$: $a{+}b \ge 1$, so $\mu_0{-}\mu_1 \le 0$ and $V_B'(1) \le 0$.

Step 5: Monotonicity from convexity. Since V_B is convex (Step 1), its derivative V_B' is non-decreasing. When $V_B'(1) \le 0$, we have $V_B'(\delta) \le V_B'(1) \le 0$ for all $\delta \in [0, 1]$, so V_B is non-increasing.

Step 6: $|B(\delta)|$ is non-decreasing. $V_W(\delta) = \mathrm{Var}(g) - V_B(\delta)$ is non-decreasing (Step 5). The function $h(x) = |\lambda|\, x/(x + \sigma_T^2)$ is strictly increasing for $x \ge 0$. Hence $|B(\delta)| = h(V_W(\delta))$ is non-decreasing. □

Condition on C_0. The proof requires $V'_B(1) \le 0$, which is guaranteed for $K{=}2$ when $(C_0)_{jj} \ge \frac{1}{2}$ (Step 4). For $K \ge 3$, $V'_B(1) \le 0$ holds for all g and p if and only if a certain matrix $M - W$ (depending on C_0 and p) is negative semi-definite; simple diagonal thresholds are sufficient in practice but not universal. Numerical verification across 20,000 random (C_0, p, g) triples per $K \in \{3, 5, 7\}$ with $(C_0)_{jj} \ge 1/K$ finds worst-case $|B|$ step deviations < 0.013 (Table 10), confirming ε-approximate monotonicity. A $K{=}2$ counterexample (C_0 the permutation matrix, $(C_0)_{jj}{=}0$) shows the diagonal condition is necessary: $V_B(\delta) = (\delta{-}\frac{1}{2})^2$ increases on $(\frac{1}{2}, 1]$.

General case (non-proportional effects). $|B(C(\delta))| \le \bar{B}(\delta)$ where $\bar{B}(\delta) = \sqrt{\mathbb{E}[\mathrm{Var}(g \mid A^*_\delta)]\,\mathbb{E}[\mathrm{Var}(f \mid A^*_\delta)]}\,/\,D(\delta)$ (Proposition 3). By the same convexity argument, each $\mathbb{E}[\mathrm{Var}(\cdot \mid A^*_\delta)]$ is non-decreasing in δ, and $D(\delta) = \mathbb{E}[\mathrm{Var}(T \mid A^*_\delta)]$ is non-decreasing. Extended verification (20,000 random C_0 matrices, $K \in \{3, 5, 7\}$) reveals worst-case step deviations of 0.011 (confounding), confirming approximate tightness; a strict monotonicity proof for the general case remains open.

A.5 Mediation Topology ($A =$ Mediator)

Data-generating process. $T \to A \to Y$, $T \to Y$.

$$\begin{aligned} T &\sim \mathrm{Bernoulli}(0.5), \\ A \mid T &\sim \mathrm{Categorical}\big(\mathrm{softmax}(\boldsymbol{\gamma} + \boldsymbol{\delta}\,T)\big), \\ Y &= \tau_{\mathrm{dir}}\,T + f(A) + \varepsilon_Y. \end{aligned}$$

Bias decomposition. By Lemma 1, the direct-effect estimator satisfies:

$$\mathrm{plim}\,\hat{\tau}_{\mathrm{dir}}(C) = \tau_{\mathrm{dir}} + \frac{\mathbb{E}[\mathrm{Cov}(T,\, f(A) \mid A^*)]}{D(C)}. \tag{18}$$

The key structural difference from confounding: T *causes* A, so the residual covariance $\mathrm{Cov}(T, f(A) \mid A^*)$ captures the indirect effect of T on Y through A that is not absorbed by conditioning on A^*.

Endpoint analysis. At $C = I$ ($A^*{=}A$): conditioning on A blocks the mediated path $T \to A \to Y$, so $\mathrm{Cov}(T, f(A) \mid A) = 0$ and $\mathrm{plim}\,\hat{\tau} = \tau_{\mathrm{dir}}$.

At $C = \frac{1}{K}\mathbf{1}\mathbf{1}^\top$ ($A^* \perp A$): no information about A is used, and

$$\begin{aligned} \mathbb{E}[\mathrm{Cov}(T, f(A) \mid A^*)] &= \mathrm{Cov}(T, f(A)), \\ D(C) &= \mathrm{Var}(T) = 0.25. \end{aligned}$$

The indirect effect is $\tau_{\mathrm{ind}} = \mathbb{E}[f(A) \mid T{=}1] - \mathbb{E}[f(A) \mid T{=}0]$, and

$$\mathrm{Cov}(T, f(A)) = 0.25\,\tau_{\mathrm{ind}}.$$

Hence at C uniform: $\mathrm{plim}\,\hat{\tau} = \tau_{\mathrm{dir}} + \tau_{\mathrm{ind}} = \tau_{\mathrm{total}}$.

Numerical Verification (Mediation sign preservation). $\mathrm{sign}(\mathrm{plim}\,\hat{\tau}_{\mathrm{dir}}) = \mathrm{sign}(\tau_{\mathrm{dir}})$ whenever $\mathrm{sign}(\tau_{\mathrm{dir}}) = \mathrm{sign}(\tau_{\mathrm{total}})$, where $\tau_{\mathrm{total}} = \tau_{\mathrm{dir}} + \tau_{\mathrm{ind}}$. This result is verified numerically for 100,000 independently sampled column-stochastic matrices C; a closed-form proof for general K remains open.

Proof. Let $R(C) = \mathbb{E}[\mathrm{Cov}(T, f(A) \mid A^*)]$ and $D(C) = \mathbb{E}[\mathrm{Var}(T \mid A^*)] > 0$.

Endpoints. At $C{=}I$: $f(A)$ is constant given $A{=}A^*$, so $R(I) = 0$ and $B(I) = 0$. At $C = \frac{1}{K}\mathbf{1}\mathbf{1}^\top$: $R = \mathrm{Cov}(T, f(A)) = 0.25\,\tau_{\mathrm{ind}}$ and $D = \mathrm{Var}(T) = 0.25$, giving $B = \tau_{\mathrm{ind}}$.

Within-group structure for binary T. Since $T \in \{0, 1\}$, the within-A^*-group bias admits a revealing decomposition. Let $p_j = P(T{=}1 \mid A^*{=}j)$. Then $\mathrm{Var}(T \mid A^*{=}j) = p_j(1{-}p_j)$ and

$$\mathrm{Cov}(T, f(A) \mid A^*{=}j) = p_j(1{-}p_j)\big[\mathbb{E}[f(A) \mid T{=}1, A^*{=}j] - \mathbb{E}[f($$

Define $\delta_j = \mathbb{E}[f(A) \mid T{=}1, A^*{=}j] - \mathbb{E}[f(A) \mid T{=}0, A^*{=}j]$ as the *conditional indirect effect* in the $A^*{=}j$ subpopulation. Then

$$B(C) = \frac{R(C)}{D(C)} = \frac{\sum_j q_j\, p_j(1{-}p_j)\,\delta_j}{\sum_j q_j\, p_j(1{-}p_j)} = \sum_j w_j\,\delta_j,$$

a weighted average of the conditional indirect effects, with weights $w_j = q_j p_j(1{-}p_j) / \sum_\ell q_\ell p_\ell(1{-}p_\ell) \ge 0$ summing to 1.

At $C{=}I$: each $A^*{=}j$ group has $A{=}j$ a.s., so $\delta_j = f(j) - f(j) = 0$ and $B = 0$. At C uniform: $\delta_j = \tau_{\mathrm{ind}}$ for all j and $B = \tau_{\mathrm{ind}}$.

Sign preservation. When $\mathrm{Cov}(g(A), f(A)) > 0$ (the indirect effect reinforces the direct effect, i.e., τ_{dir} and τ_{ind} have the same sign), the bias $B(C) \ge 0$ cannot oppose τ_{dir}, so sign is preserved trivially.

When τ_{dir} and τ_{ind} have opposite signs but $|\tau_{\text{ind}}| < |\tau_{\text{dir}}|$ (i.e., $\text{sign}(\tau_{\text{dir}}) = \text{sign}(\tau_{\text{total}})$), sign reversal requires some $|\delta_j| > |\tau_{\text{dir}}|$ with the opposing sign. Since each δ_j is a difference of weighted averages of f values (with weights $P(A{=}k \mid T{=}t, A^*{=}j)$), and $B(C)$ is a convex combination of δ_j, the bound $|B(C)| \leq \max_j |\delta_j|$ holds. We verify numerically that $\max_j |\delta_j| \leq |\tau_{\text{ind}}|$ for all 100,000 random C matrices in our experiments, confirming sign preservation. □

Verification for paper parameters ($K{=}3$). With $\tau_{\text{dir}}{=}0.5$, $\boldsymbol{\gamma}{=}(0,0,0)$, $\boldsymbol{\delta}{=}(0, 0.8, -0.6)$, $\boldsymbol{\beta}{=}(0, 1.0, -0.5)$:

$\tau_{\text{ind}} = \mathbb{E}[f(A) \mid T{=}1] - \mathbb{E}[f(A) \mid T{=}0] = 0.351$ (computed from the softmax class probabilities). $\tau_{\text{total}} = 0.5 + 0.351 = 0.851 > 0$. Since $\tau_{\text{dir}}{>}0$ and $\tau_{\text{total}}{>}0$, sign preservation holds for all C.

Mediation monotonicity (proportional effects). Under the interpolation $C(\delta) = (1{-}\delta)I + \delta C_0$, we prove $|B(C(\delta))|$ is non-decreasing in δ when the mediator's effect on Y is proportional to its informativeness about T: $f(A) = \lambda\, g(A)$ for some scalar λ, where $g(k) = \mathbb{E}[T \mid A{=}k]$.

Step 1: Decompose T via the mediator. Define $g(A) = \mathbb{E}[T \mid A]$ and $\varepsilon_T = T - g(A)$. By construction, $\mathbb{E}[\varepsilon_T \mid A] = 0$. By non-differential misclassification ($T \perp A^* \mid A$): $\mathbb{E}[\varepsilon_T \mid A{=}k, A^*{=}j] = 0$, so $\mathbb{E}[\varepsilon_T \mid A^*{=}j] = 0$ and $\text{Cov}(\varepsilon_T, g(A) \mid A^*{=}j) = 0$ for every j.

Step 2: Reduce the numerator. Since $f = \lambda g$:

$$R(C) = \lambda\, \mathbb{E}[\text{Var}(g(A) \mid A^*)] = \lambda\, V_W.$$

Step 3: Constant additive term in denominator.

$$D(C) = \mathbb{E}[\text{Var}(T \mid A^*)] = V_W + \sigma_\varepsilon^2,$$

where $\sigma_\varepsilon^2 \equiv \mathbb{E}[\text{Var}(\varepsilon_T \mid A^*)]$. This is *constant* (independent of C): $\mathbb{E}[\text{Var}(\varepsilon_T \mid A^*)] = \sum_k p_k\, \text{Var}(T \mid A{=}k) = \mathbb{E}[\text{Var}(T \mid A)]$, where the equality uses column-stochasticity: $\sum_j q_j\, P(A{=}k \mid A^*{=}j) = p_k$.

Step 4: Closed form and monotonicity. $B(C) = \lambda\, V_W/(V_W + \sigma_\varepsilon^2)$, structurally identical to confounding (Proposition 2) with σ_ε^2 playing the role of σ_T^2. Since $V_B(\delta)$ is convex with $V_B'(0) \leq 0$ (same argument as §A.4, Steps 1–3), $V_W(\delta) = V - V_B(\delta)$ is non-decreasing under the same conditions on C_0. Since $h(x) = |\lambda| x/(x + \sigma_\varepsilon^2)$ is increasing, $|B(C(\delta))|$ is non-decreasing.

Boundedness of the leakage coefficient. The mediation bias (3) writes the residual indirect effect as $(1 - \lambda(C))\, \tau_{\text{ind}}$, which (for $\tau_{\text{ind}} \neq 0$) *implicitly* defines

$$\lambda(C) \;:=\; 1 - \frac{B(C)}{\tau_{\text{ind}}}, \qquad B(C) \;\equiv\; \operatorname{plim} \hat{\tau}_{\text{dir}}(A^*) - \tau_{\text{dir}}, \qquad \tau_{\text{ind}} = \tag{19}$$

Substituting back into (3) gives the convex-combination form $\operatorname{plim} \hat{\tau}_{\text{dir}}(C) = \lambda(C)\, \tau_{\text{dir}} + (1 - \lambda(C))\, \tau_{\text{total}}$, so $\lambda(C)$ is the weight placing the biased plim between τ_{dir} (perfect classification) and τ_{total} (no information).

Proposition 4 (Mediation leakage boundedness)**.** *Let $\lambda(C)$ be defined by (19) with $\tau_{\text{ind}} \neq 0$, and let C be column-stochastic.*

(a) $\lambda(C) \leq 1$ iff $\text{sign}(B(C)) = \text{sign}(\tau_{\text{ind}})$ (the bias points in the direction of the indirect effect; no overcorrection).

(b) $\lambda(C) \geq 0$ iff additionally $|B(C)| \leq |\tau_{\text{ind}}|$ (the bias does not exceed the indirect effect in magnitude).

Equivalently, $\lambda(C) \in [0,1]$ iff $\operatorname{plim} \hat{\tau}_{\text{dir}}(C) \in [\min(\tau_{\text{dir}}, \tau_{\text{total}}), \max(\tau_{\text{dir}}, \tau_{\text{total}})]$.

Under proportional effects *($f(A) = \lambda_0\, g(A)$ with $g(k) = \mathbb{E}[T \mid A{=}k]$), both (a) and (b) hold for every C, so $\lambda(C) \in [0,1]$ unconditionally. For* general (non-proportional) *f, both (a) and (b) can fail because $\text{Cov}(\mathbb{E}[g \mid A^*], \mathbb{E}[f \mid A^*])$ may differ in sign or magnitude from $\text{Cov}(g, f)$ (a covariance-level analogue of Simpson's paradox under A^* aggregation); the numerical verification below quantifies the empirical rate. The endpoints $\lambda(I) = 1$ and $\lambda(C) \to 0$ in the rank-one (uninformative) limit hold in all regimes.*

Sign preservation. *When $\lambda(C) \in [0,1]$, the convex-combination form gives $\operatorname{plim} \hat{\tau}_{\text{dir}}(C) \in [\min(\tau_{\text{dir}}, \tau_{\text{total}}), \max(\tau_{\text{dir}}, \tau_{\text{total}})]$. If additionally $\text{sign}(\tau_{\text{dir}}) = \text{sign}(\tau_{\text{total}})$, both endpoints share the same sign, so $\text{sign}(\operatorname{plim} \hat{\tau}_{\text{dir}}) = \text{sign}(\tau_{\text{dir}})$ for every C. Under proportional effects, $\lambda(C) \in [0,1]$ is automatic (above), so sign preservation of $\hat{\tau}_{\text{dir}}$ reduces to the single condition $\text{sign}(\tau_{\text{dir}}) = \text{sign}(\tau_{\text{total}})$.*

Proof. (a),(b). With $\tau_{\text{ind}} \neq 0$, $\lambda(C) \leq 1 \iff B(C)/\tau_{\text{ind}} \geq 0 \iff \operatorname{sign}(B(C)) = \operatorname{sign}(\tau_{\text{ind}})$. Given (a), $\lambda(C) \geq 0 \iff B(C)/\tau_{\text{ind}} \leq 1 \iff |B(C)| \leq |\tau_{\text{ind}}|$.

Proportional case. Step 4 of the mediation derivation gives $B(C) = \lambda_0 \, V_W(C)/(V_W(C) + \sigma_\varepsilon^2)$; the rank-one limit $V_W \to V$ yields $\tau_{\text{ind}} = \lambda_0 \, V/(V + \sigma_\varepsilon^2)$. Both quantities carry $\operatorname{sign}(\lambda_0)$, establishing (a). The ratio

$$\frac{B(C)}{\tau_{\text{ind}}} = \frac{V_W(C)\,(V + \sigma_\varepsilon^2)}{V\,(V_W(C) + \sigma_\varepsilon^2)}$$

is monotone increasing in $V_W(C) \in [0, V]$ with range $[0, 1]$ (its derivative in V_W is $\sigma_\varepsilon^2(V + \sigma_\varepsilon^2)/[V(V_W + \sigma_\varepsilon^2)^2] > 0$), establishing (b). Hence $\lambda(C) \in [0, 1]$ unconditionally.

General case. Substituting $\lambda(C)$ from (19) into (3) yields $\operatorname{plim} \hat{\tau}_{\text{dir}}(C) = \lambda(C)\,\tau_{\text{dir}} + (1-\lambda(C))\,\tau_{\text{total}}$, so $\lambda(C) \in [0, 1]$ iff $\operatorname{plim} \hat{\tau}_{\text{dir}}(C)$ lies in the closed interval with endpoints τ_{dir} and τ_{total}.

Endpoints. At $C{=}I$, $A^*{=}A$ a.s., $\mathbb{E}[T \mid A^*] = g(A)$, and FWL gives $\operatorname{plim} \hat{\tau}_{\text{dir}} = \tau_{\text{dir}}$, hence $B(I) = 0$ and $\lambda(I) = 1$. At any rank-one C (rows identical), $A^* \perp A$ so $\mathbb{E}[T \mid A^*]$ is constant, the partial regression on A^* has no effect, and $\operatorname{plim} \hat{\tau}_{\text{dir}} = \operatorname{Cov}(Y, T)/\operatorname{Var}(T) = \tau_{\text{total}}$, giving $B = \tau_{\text{ind}}$ and $\lambda = 0$. ☐

Relation to the between-variance ratio. A natural *diagnostic* closely related to $\lambda(C)$ is the between-variance ratio

$$\lambda_V(C) := \frac{V_B(C)}{V}, \qquad V_B(C) = \operatorname{Var}\big(\mathbb{E}[g(A) \mid A^*]\big), \quad V = \operatorname{Var}(g(A)), \tag{20}$$

which satisfies $\lambda_V(C) \in [0, 1]$ *unconditionally* by the law of total variance applied to $g(A)$. Under proportional effects, λ and λ_V agree at the endpoints and are related by

$$1-\lambda(C) = \frac{V_W(C)\,(V + \sigma_\varepsilon^2)}{V\,(V_W(C) + \sigma_\varepsilon^2)} = (1-\lambda_V(C)) \cdot \frac{V + \sigma_\varepsilon^2}{V_W(C) + \sigma_\varepsilon^2},$$

so they coincide in the limit $\sigma_\varepsilon^2 \gg V$. They generally differ for non-proportional f. We use λ throughout the main text because it parameterizes (3) directly; λ_V remains a useful summary of the channel quality of A^* as a proxy for A (independent of the structural form of f).

Numerical verification. For each $K \in \{3, 5, 7\}$ we draw $N{=}10{,}000$ configurations $(\pi, g, \sigma_\varepsilon^2, C)$ with $\pi \sim \operatorname{Dirichlet}(\mathbf{1})$, $g \sim \mathcal{N}(0, I_K)$, $\sigma_\varepsilon^2 \sim \operatorname{Uniform}(0.1, 2)$, columns of $C \sim \operatorname{Dirichlet}(\mathbf{1})$, and evaluate $\lambda(C)$ from (19) in two regimes for f:

- **Proportional** ($f = \lambda_0\, g$, $\lambda_0 \sim \mathcal{N}(0,1) \setminus \{0\}$): $\lambda(C) \in [0, 1]$ in *all* 30,000 trials (pooled observed range $[1.2{\times}10^{-5}, 0.850]$; zero out-of-bounds), confirming the closed-form bound. Sign preservation of $\hat{\tau}_{\text{dir}}$ therefore reduces to $\operatorname{sign}(\tau_{\text{dir}}) = \operatorname{sign}(\tau_{\text{total}})$.
- **Non-proportional** ($f \sim \mathcal{N}(0, I_K)$ drawn independently of g, restricted to $|\tau_{\text{ind}}| \geq 0.05$ to avoid the ill-conditioned regime where $\lambda{=}1{-}B/\tau_{\text{ind}}$ diverges): $\lambda(C)$ exceeds $[0, 1]$ in $\sim$ 15–17% of trials, illustrating that (a) and (b) are not automatic outside the proportional case.

The diagnostic $\lambda_V(C)$ lies in $[0, 1]$ across all 30,000 trials irrespective of regime, as guaranteed by the law of total variance. The script and full results are in `artifacts/verify_lambda_bound.py` and `artifacts/verify_lambda_bound_results.json`.

A.6 Collider Topology (A = Collider)

Data-generating process. $T \to A \leftarrow Y$, $T \to Y$. The treatment effect is unconfounded:

$$T \sim \mathcal{N}(0,1), \quad Y = \tau\, T + \varepsilon_Y, \quad \varepsilon_Y \sim \mathcal{N}(0,1), \; \varepsilon_Y \perp T,$$
$$A \mid T, Y \sim \operatorname{Categorical}\big(\operatorname{softmax}(\boldsymbol{\delta}_T\, T + \boldsymbol{\delta}_Y\, Y)\big).$$

The collider A is a common *effect* of T and Y. The no-control estimator (regressing Y on T alone) is consistent: $\operatorname{plim} \hat{\tau}_{\text{nc}} = \tau$. The researcher erroneously controls for A^*, inducing Berkson's bias.

Bias decomposition. Unlike confounding, ε_Y is not independent of A (since A depends on $Y = \tau T + \varepsilon_Y$), so Lemma 1 does not apply directly. Instead, we derive the bias from the FWL identity (7). Writing $\tilde{Y} = \tau \tilde{T} + (\varepsilon_Y - \mathbb{E}[\varepsilon_Y \mid A^*])$ and noting that $\mathbb{E}[\tilde{T}\, \varepsilon_Y] = \mathbb{E}[\operatorname{Cov}(T, \varepsilon_Y \mid A^*)]$ (since $T \perp \varepsilon_Y$ unconditionally

ally):

$$\operatorname{plim}\hat{\tau}(C) = \tau + B(C), \quad B(C) = \frac{\mathbb{E}[\operatorname{Cov}(T,\, \varepsilon_Y \mid A^*)]}{\mathbb{E}[\operatorname{Var}(T \mid A^*)]} \tag{21}$$

Under non-differential misclassification ($A^* \perp (T, \varepsilon_Y) \mid A$), define $g(k) = \mathbb{E}[T \mid A{=}k]$ and $h(k) = \mathbb{E}[\varepsilon_Y \mid A{=}k]$. By the tower property, $\mathbb{E}[T \mid A^*] = \mathbb{E}[g(A) \mid A^*]$ and $\mathbb{E}[\varepsilon_Y \mid A^*] = \mathbb{E}[h(A) \mid A^*]$. Since $\operatorname{Cov}(T, \varepsilon_Y) = 0$, the law of total covariance gives:

$$B(C) = \frac{-\operatorname{Cov}\big(\mathbb{E}[g(A) \mid A^*],\, \mathbb{E}[h(A) \mid A^*]\big)}{\mathbb{E}[\operatorname{Var}(T \mid A^*)]}.$$

Endpoint analysis. At $C = I$ ($A^*{=}A$): $B(I) = -\operatorname{Cov}(g(A), h(A)) \,/\, \mathbb{E}[\operatorname{Var}(T \mid A)] = B_{\text{Berkson}}$, the full Berkson's bias from conditioning on the collider.

At $C = \frac{1}{K}\mathbf{1}\mathbf{1}^\top$ ($A^* \perp A$): $\mathbb{E}[g(A) \mid A^*]$ and $\mathbb{E}[h(A) \mid A^*]$ are constant, so $B = 0$ and $\operatorname{plim}\hat{\tau} = \tau$.

This is the structural reverse of confounding: misclassification attenuates the collider-induced bias toward zero, rather than growing bias from zero toward B_{OV}.

Proposition 5 (Collider sign preservation — proportional case)**.** *If $h(A) = \lambda\, g(A)$ for some constant λ, then $|B(C)| \le |B(I)|$ for all column-stochastic C, and $B(C)$ has the same sign as $B(I)$. Consequently,* $\operatorname{sign}(\operatorname{plim}\hat{\tau}) = \operatorname{sign}(\tau)$ *whenever* $\operatorname{sign}(\tau) = \operatorname{sign}(\tau + B_{\text{Berkson}})$.

Proof. When $h = \lambda g$, the numerator becomes $-\lambda \operatorname{Var}(\mathbb{E}[g(A) \mid A^*])$ and the denominator is $\operatorname{Var}(T) - \operatorname{Var}(\mathbb{E}[g(A) \mid A^*])$. Let $V^* = \operatorname{Var}(\mathbb{E}[g(A) \mid A^*]) \in [0, V_g]$ where $V_g = \operatorname{Var}(g(A))$. Then

$$|B(C)| = |\lambda| \,\frac{V^*}{\operatorname{Var}(T) - V^*}.$$

The function $x \mapsto x/(a - x)$ is increasing on $[0, a)$. With $a = \operatorname{Var}(T) > V_g \ge V^*$: $|B(C)| \le |\lambda|\, V_g/(\operatorname{Var}(T) - V_g) = |B(I)|$. At $V^*{=}0$ (uniform C): $B{=}0$. At $V^*{=}V_g$ ($C{=}I$): $B{=}B(I)$. □

Proposition 6 (Collider sign preservation — general case)**.** *Define*

$$\bar{B}_{\text{col}} = \frac{\sqrt{\operatorname{Var}(g(A))\, \operatorname{Var}(h(A))}}{\mathbb{E}[\operatorname{Var}(T \mid A)]}. \tag{22}$$

If $|\tau| > \bar{B}_{\text{col}}$, then $\operatorname{sign}(\operatorname{plim}\hat{\tau}(C)) = \operatorname{sign}(\tau)$ *for all column-stochastic C.*

Proof. Step 1: Bound the numerator. By Cauchy–Schwarz: $|\operatorname{Cov}(\mathbb{E}[g \mid A^*], \mathbb{E}[h \mid A^*])| \le \sqrt{\operatorname{Var}(\mathbb{E}[g \mid A^*])\, \operatorname{Var}(\mathbb{E}[h \mid A^*])}$. Since conditional expectation reduces variance: $\operatorname{Var}(\mathbb{E}[g \mid A^*]) \le \operatorname{Var}(g(A))$ and likewise for h.

Step 2: Bound the denominator. $\mathbb{E}[\operatorname{Var}(T \mid A^*)] = \operatorname{Var}(T) - \operatorname{Var}(\mathbb{E}[T \mid A^*]) \ge \operatorname{Var}(T) - \operatorname{Var}(g(A)) = \mathbb{E}[\operatorname{Var}(T \mid A)]$.

Step 3: Conclude. $|B(C)| \le \sqrt{\operatorname{Var}(g)\, \operatorname{Var}(h)} \,/\, \mathbb{E}[\operatorname{Var}(T \mid A)] = \bar{B}_{\text{col}}$. When $|\tau| > \bar{B}_{\text{col}}$, the bias cannot overwhelm τ. In practice, $\operatorname{Var}(g(A))$ and $\operatorname{Var}(h(A))$ can be estimated from observable data by bootstrapping regression coefficients from A^*-stratified subsamples, yielding conservative upper bounds on $\bar{B}_{\text{col}}$. □

Verification for paper parameters (K=3). With $\tau{=}1.0$, $\boldsymbol{\delta}_T{=}(0,\, 0.5,\, -0.3)$, $\boldsymbol{\delta}_Y{=}(0,\, 0.8,\, 0.6)$, from $5{\times}10^6$-sample Monte Carlo:

$$\operatorname{Var}(g(A)) = 0.187, \quad \operatorname{Var}(h(A)) = 0.076, \quad \mathbb{E}[\operatorname{Var}(T \mid A)] = 0$$
$$\bar{B}_{\text{col}} = \sqrt{0.187 \times 0.076} \,/\, 0.813 = 0.147.$$

Since $|\tau|{=}1.0 \gg \bar{B}_{\text{col}}{=}0.147$, Proposition 6 guarantees sign preservation. Additionally, $B_{\text{Berkson}}{=}{-0.114}$, giving $\tau + B_{\text{Berkson}}{=}0.886{>}0$. Verified across 100,000 random column-stochastic C: $\tau_{\text{sign flip}}{=}0\%$, and $|B(C)| \le |B_{\text{Berkson}}|$ in all cases.

A.7 Front-door Topology (A = Mediator under Confounding)

Data-generating process. $T \to A \to Y$, $U \to T$, $U \to Y$. An unobserved confounder U affects both treatment and outcome:

$$U \sim \mathcal{N}(0,1), \quad T = \alpha_U\, U + \eta, \quad \eta \sim \mathcal{N}(0,1),\ \eta \perp U,$$
$$A \mid T \sim \operatorname{Categorical}\big(\operatorname{softmax}(\boldsymbol{\gamma} + \boldsymbol{\delta}\, T)\big),$$
$$Y = f(A) + \lambda\, U + \varepsilon_Y, \quad \varepsilon_Y \perp (T, A, U).$$

There is no direct $T{\to}Y$ edge; the causal effect of T on Y operates entirely through A. The researcher controls for A^* in $Y \sim 1 + T + D_1^* + \cdots + D_{K-1}^*$.

Bias decomposition. Since the direct effect is zero, the FWL coefficient on T captures only bias:

$$\operatorname{plim}\hat{\tau}(C) = \frac{\mathbb{E}[\operatorname{Cov}(T,\, Y \mid A^*)]}{\mathbb{E}[\operatorname{Var}(T \mid A^*)]}.$$

Substituting $Y = f(A) + \lambda U + \varepsilon_Y$ and using $\varepsilon_Y \perp (T, A, A^*)$:

$$\operatorname{plim} \hat{\tau}(C) = \underbrace{\frac{\lambda\,\alpha_U}{\operatorname{Var}(T)}}_{\text{confounding (constant)}} + \underbrace{\frac{\mathbb{E}[\operatorname{Cov}(T,\, f(A) \mid A^*)]}{D(C)}}_{B_{\text{med}}(C)}, \tag{23}$$

where $D(C) = \mathbb{E}[\operatorname{Var}(T \mid A^*)]$.

Derivation of (23). Expand the numerator as $\mathbb{E}[\operatorname{Cov}(T, f(A) \mid A^*)] + \lambda\,\mathbb{E}[\operatorname{Cov}(T, U \mid A^*)]$. For the confounding term: since $T = \alpha_U U + \eta$ with (U, η) jointly Gaussian, the regression residual $\xi = U - \frac{\alpha_U}{\operatorname{Var}(T)} T$ satisfies $\xi \perp T$. Because A is a function of T (plus independent softmax noise) and A^* is a function of A (plus independent misclassification noise), $\xi \perp (T, A, A^*)$. Therefore

$$\operatorname{Cov}(T, U \mid A^*{=}j) = \frac{\alpha_U}{\operatorname{Var}(T)} \operatorname{Var}(T \mid A^*{=}j) \quad \forall\, j,$$

and $\lambda\,\mathbb{E}[\operatorname{Cov}(T, U \mid A^*)] = \frac{\lambda\alpha_U}{\operatorname{Var}(T)} D(C)$, yielding the constant first term after division by $D(C)$. □

This decomposition reveals that *misclassification does not affect the confounding component.* The confusion matrix C modulates only the mediation-type leakage $B_{\text{med}}(C)$, which has the same structure as in §A.5.

Endpoint analysis. At $C = I$: $B_{\text{med}}(I) = 0$ (controlling A blocks the $T \to A \to Y$ path), and $\operatorname{plim} \hat{\tau}(I) = \lambda\alpha_U / \operatorname{Var}(T)$ (pure confounding residual).

At $C = \frac{1}{K}\mathbf{1}\mathbf{1}^\top$: $B_{\text{med}} = \operatorname{Cov}(T, f(A)) / \operatorname{Var}(T)$, and $\operatorname{plim} \hat{\tau}_{\text{nc}} = [\operatorname{Cov}(T, f(A)) + \lambda\alpha_U] / \operatorname{Var}(T)$, the naive regression coefficient with no controls.

Numerical Verification (Front-door sign preservation). Since $B_{\text{med}}(C)$ has the same structure as the mediation bias (§A.5)—interpolating between 0 at $C{=}I$ and $\operatorname{Cov}(T, f(A)) / \operatorname{Var}(T)$ at C uniform—sign preservation holds whenever the two endpoints have the same sign:

$$\operatorname{sign}\big(\lambda\alpha_U\big) = \operatorname{sign}\big(\lambda\alpha_U + \operatorname{Cov}(T, f(A))\big).$$

This result is verified numerically for 100,000 independently sampled column-stochastic matrices C.

Verification for paper parameters ($K{=}3$). With $\alpha_U{=}1.0$, $\lambda{=}1.0$, $\boldsymbol{\beta}{=}(0, 1.0, -0.5)$, $\boldsymbol{\gamma}{=}(0, 0.5, -0.3)$, $\boldsymbol{\delta}{=}(0, 1.0, -0.8)$:

$$\lambda\alpha_U / \operatorname{Var}(T) = 1.0 \times 1.0 \,/\, 2.0 = 0.500,$$
$$\operatorname{Cov}(T, f(A)) / \operatorname{Var}(T) = 0.594 \,/\, 2.0 = 0.297.$$

Both components are positive, so $\operatorname{plim} \hat{\tau}(C) \in [0.500,\, 0.797] > 0$ for all C.

A.8 Exposure Topology: Sign Reversal Counterexample

Proposition 7 (Exposure sign reversal)**.** *For $K{=}3$ with A as the treatment variable, there exist a marginal distribution $\mathbf{p}$, treatment effects $\boldsymbol{\beta}$, and a column-stochastic C with $C_{kk} > 0.5$ for all k, such that* $\operatorname{sign}(\operatorname{plim} \hat{\beta}_1) \neq \operatorname{sign}(\beta_1)$.

Construction. Consider the exposure DGP $Y = \beta_1 D_1 + \beta_2 D_2 + \varepsilon_Y$ with:

- Uniform prior: $\mathbf{p} = (1/3,\, 1/3,\, 1/3)$.
- True effects: $\beta_1 = 1,\ \beta_2 = -10$.

Choose the confusion matrix

$$C = \begin{pmatrix} 0.60 & 0.10 & 0.10 \\ 0.10 & 0.60 & 0.39 \\ 0.30 & 0.30 & 0.51 \end{pmatrix}, \qquad C_{kk} \in \{0.60,\, 0.60,\, 0.51\}, \text{ all >}$$

Column sums are 1 (verifiable by inspection).

The OLS estimator regresses Y on $(1, D_1^*, D_2^*)$. The plim of $\hat{\beta}_j$ equals $\mathbb{E}[Y \mid A^*{=}j] - \mathbb{E}[Y \mid A^*{=}0]$, where

$$\mathbb{E}[Y \mid A^*{=}j] = \sum_{k=0}^{2} P(A{=}k \mid A^*{=}j)\,\beta_k = \sum_{k=1}^{2} \frac{C_{jk}\, p_k}{q_j}\,\beta_k.$$

Marginal distribution of A^*:

$$q_0 = \tfrac{1}{3}(0.60 + 0.10 + 0.10) = 0.267,$$
$$q_1 = \tfrac{1}{3}(0.10 + 0.60 + 0.39) = 0.363,$$
$$q_2 = \tfrac{1}{3}(0.30 + 0.30 + 0.51) = 0.370.$$

Conditional means:

$$\mathbb{E}[Y \mid A^*{=}0] = \frac{0.10 \times \frac{1}{3} \times 1 + 0.10 \times \frac{1}{3} \times (-10)}{0.267} = \frac{0.0333 - \text{(}}{0.267}$$

$$\mathbb{E}[Y \mid A^*{=}1] = \frac{0.60 \times \frac{1}{3} \times 1 + 0.39 \times \frac{1}{3} \times (-10)}{0.363} = \frac{0.200 - 1.}{0.363}$$

$$\mathbb{E}[Y \mid A^*{=}2] = \frac{0.30 \times \frac{1}{3} \times 1 + 0.51 \times \frac{1}{3} \times (-10)}{0.370} = \frac{0.100 - 1.}{0.370}$$

Estimated coefficient:

$$\text{plim}\,\hat{\beta}_1 = \mathbb{E}[Y \mid A^*{=}1] - \mathbb{E}[Y \mid A^*{=}0] = -3.028 - (-1.125) = \mathbf{-1.903}.$$

The true $\beta_1 = +1 > 0$ but $\text{plim}\,\hat{\beta}_1 = -1.903 < 0$: **sign reversal**.

Mechanism. The large off-diagonal entry $C_{12} = 0.39$ (probability of classifying true category 2 as $A^*{=}1$) routes substantial probability mass from category 2 ($\beta_2 = -10$) into the $A^*{=}1$ bin. Despite $C_{11} = 0.60 > 0.5$, the contamination from the strongly negative β_2 overwhelms the positive β_1, reversing the sign of the estimated effect for category 1. This failure mode is structurally impossible at $K{=}2$: a $2{\times}2$ column-stochastic matrix with $C_{kk} > 0.5$ has the form $C = \begin{pmatrix} a & 1-b \\ 1-a & b \end{pmatrix}$ with $a, b > 0.5$, and the single coefficient $\hat{\beta}_1$ is a positive-weighted average of β_1, preserving sign. For $K{\geq}3$, the additional degrees of freedom in C permit cross-contamination between categories with opposite-sign effects. □

A.9 Summary of Conditions

Topology	**Sufficient condition**	**Status**	**Paper DGP**
Confounding	$\text{sign}(\tau) = \text{sign}(\tau + B_{\text{OV}})$	Proved	✓ (2.0 vs 2.3)
Collider	$\lvert\tau\rvert > \bar{B}_{\text{col}}$ (Prop. 6)	Proved	✓ (1.0 vs 0.15)
Mediation	$\text{sign}(\tau_{\text{dir}}) = \text{sign}(\tau_{\text{total}})$	Empirical	✓ (0.3 vs 0.85)
M-bias	Bias is attenuating (toward τ)	Empirical	✓
Front-door	$\text{sign}(\lambda\alpha_U) = \text{sign}(\lambda\alpha_U + \text{Cov}(T, f))$	Empirical	✓ (0.50 vs 0.80)
Exposure	Not guaranteed for $K \geq 3$	Counterex.	See §A.8
IV	Not guaranteed (denominator instability)	—	—

Table 2: Sufficient conditions for sign preservation under non-differential misclassification. "Proved" = algebraic proof via Cauchy–Schwarz (confounding) or Berkson attenuation (collider). "Empirical" = verified across 100,000 random column-stochastic matrices with zero violations; analytic proof for general K remains open. All conditions are on DGP parameters (not on C): once satisfied, sign preservation holds for all column-stochastic C with $C_{kk} > 0$.

Structural argument for M-bias topology. *M-bias:* The M-structure creates a backdoor path through two latent common causes. Misclassification of A weakens the statistical association along this backdoor path, reducing the spurious correlation rather than amplifying it. The bias therefore interpolates between the full M-bias (perfect classification, $\delta{=}0$) and zero (no classification information, $\delta{\to}\delta_{\max}$), monotonically non-increasing in δ.

M-bias sign-flip ASV $= \infty$. For the four covariate topologies (confounding, mediation, collider, M-bias), FWL invariance (Appendix A.2) implies that the coefficient on T depends on A^* only through the residualized relationship $\tilde{T} = T - \mathbb{E}[T \mid A^*]$. For M-bias, the spurious path through A is the sole source of bias, and misclassification attenuates this path for all column-stochastic C. Since $\text{bias}(C(\delta))$ is monotonically non-increasing in δ and starts at a value with the same sign as τ, no $\delta \in [0, \delta_{\max}]$ can produce a sign reversal. Therefore $\delta^*_{\text{sign}} = \infty$, which is a direct consequence of FWL invariance combined with the protective monotonicity of M-bias misclassification.

Remark. The sign preservation conditions for confounding, mediation, collider, and front-door are conditions on the DGP parameters (the relative strength of the causal effect versus the confounding, indirect, or collider-induced bias), not on the confusion matrix C. Once the condition is satisfied, sign preservation holds for *all* column-stochastic C with $C_{kk} > 0$—no diagonal-dominance restriction is needed. Conversely, if the condition fails (i.e., the omitted-variable or total-effect estimate has the opposite sign from τ), then there exist confusion matrices that produce sign reversal. For the DGP parameters used throughout our experiments, the conditions are comfortably satisfied. The sign-alignment condition is violated when the confounder has opposing effects on treatment and outcome (e.g., $\gamma_A > 0$ on T but $\gamma_A < 0$ on Y), producing $\text{sign}(\tau) \neq \text{sign}(\tau_{\text{OV}})$. In applied settings, such opposing effects are uncommon when the analyst adjusts for A precisely because they believe it confounds the T–Y relationship in a consistent direction. When the condition fails, sign reversal becomes possible under sufficient misclassification, though this represents a pathological DGP where adjustment itself may be questionable.

B Supplementary Tables, Figures, and Algorithm

B.1 Binarization Collapse

Collapsing K-class annotations to binary before analysis discards multi-class error structure. We quantify "binarization collapse" as

$\Delta = |\text{bias}_K - \text{bias}_{K\to 2}|$.

For K=3, binarization introduces additional bias varying 154× across topologies: from $\bar{\Delta}$=0.006 (M-bias) to $\bar{\Delta}$=0.926 (IV). Exposure and IV also introduce sign flips during binarization (14% and 17.5%), a phenomenon absent in the other topologies (Figure 6).

Table 3: ASV type applicability and sensitivity across DAG topologies. Magnitude ASV at τ=0.5, 10% threshold. Power at τ=0.05 significance filter. —: not applicable.

Topology	Mag. ASV	Signif.	Sign-flip	Rank
IV	0.101–0.607	0%	1.1–6.7	1 (fragile)
Front-door	0.064–0.387	100%	—	2
Confounding	0.081–0.487	100%	—	3
Mediation	0.076–0.454	0%	—	4
Collider	0.338–2.028	0%	—	5
M-bias	∞	96–100%	—	6 (equivalent)
Exposure	—	—	83.4%*	—

*Fraction of C_0 with $\tau_{\text{sign\_flip}} < 1$. Exposure uses sign-flip ASV exclusively.

Table 4: RMSE reduction (%) by correction method and DAG topology (mean across 6 error structures, N_{sim}=1,000). Positive = reduced RMSE; negative = increased RMSE. Naive/MLA show 0.0% for FWL-invariant topologies because C^{-1} cannot change the T coefficient. **Bold**: best method per topology; underline: best gold-free method.

Topology	Naive	MLA	DSL	SIMEX	PPI++
Confounding	0.0	0.0	48.0	35.1	**83.2**
Mediation	0.0	0.0	52.9	39.1	**54.0**
Collider	0.0	0.0	−92.6	31.8	**50.5**
M-bias	0.0	0.0	−267	**0.1**	−88.0
Front-door	−52.7	−111.5	−29.0	−62.4	**74.1**
Exposure	−44.8	−89.0	−59.5	−9.8	**75.5**
IV	−40.4	−58.7	−206.1	59.4	**79.4**
Grand mean	−19.7	−37.0	−79.0	13.3	**47.0**

Error structures used in correction experiments. The six confusion matrix structures used for Table 4 are: (1) *Diagonal-dominant*: C_{kk}=0.7, off-diagonal uniform; (2) *Moderate noise*: C_{kk}=0.5, off-diagonal uniform; (3) *Near-permutation*: C_{kk}=0.4, largest off-diagonal 0.5; (4) *Asymmetric*: C_{kk} varies (0.8, 0.5, 0.3); (5) *Block confusion*: two classes confused (C_{12}=C_{21}=0.4), third clean; (6) *Heavy noise*: C_{kk}=0.4, off-diagonal uniform. All are column-stochastic with K=3. Results are averaged across all six structures.

B.2 Proof of FWL Invariance

For the four topologies where A enters as a covariate (confounding, mediation, collider, M-bias), the regression is $Y = \tau T + \boldsymbol{\gamma}^\top \mathbf{D}^* + \varepsilon$, where $\mathbf{D}^*$ is the $N \times (K-1)$ dummy matrix for A^*. Any matrix-based correction replaces $\mathbf{D}^*$ with $\tilde{\mathbf{D}} = \mathbf{D}^* M$ for some invertible $(K-1) \times (K-1)$ matrix M (e.g., $M = C^{-1}$ for plug-in, or $M = (C^\top C)^{-1} C^\top$ for MLE). By the Frisch–Waugh–Lovell theorem (Lovell, 1963), $\hat{\tau}$ depends only on the residual of T projected off $\text{col}(\mathbf{D}^*)$. Since $\text{col}(\mathbf{D}^* M) = \text{col}(\mathbf{D}^*)$ for invertible M, the projection is unchanged and $\hat{\tau}(\tilde{\mathbf{D}}) = \hat{\tau}(\mathbf{D}^*)$ exactly. SIMEX adds noise $\mathbf{D}^*_{\text{sim}} = \mathbf{D}^* + \lambda \mathbf{E}$; when $\mathbf{E}$ is generated from the same column space, the extrapolation to $\lambda = -1$ converges to a column-space-equivalent transformation, yielding 0% bias reduction empirically.

FWL breakdown under logistic regression. The column-space argument applies only to estimators whose treatment coefficient is a function of T residualized off $\text{col}(\mathbf{D}^*)$—a property of linear projection. The logistic MLE solves $\sum_i (Y_i - \sigma(\mathbf{x}_i^\top \boldsymbol{\beta}))\mathbf{x}_i = \mathbf{0}$; the implicit weighting by $\sigma'(\cdot)$ depends on the actual values of the design columns, not only on their span, so replacing $\mathbf{D}^*$ by $\mathbf{D}^* M$ generally changes the fitted $\hat{\tau}$. We verify this on the confounding topology (K=3, τ=0.5, N=10,000 × 500 MC reps) under both linear ($Y = \tau T + \mathbf{D}\boldsymbol{\beta} + \varepsilon$) and logistic ($Y \sim \text{Bern}(\sigma(\tau T + \mathbf{D}\boldsymbol{\beta}))$) DGPs, comparing the uncorrected estimator (fit on $\mathbf{D}^*$) against Naive plug-in (fit on per-sample $(C^{-1}\mathbf{e}_{A^*})_{1:K}$). Per-rep estimates are paired (same A^*, same outcome).

Two findings: (i) Under OLS, $\hat{\tau}^{(m)}_{\text{corr}} = \hat{\tau}^{(m)}_{\text{un}}$ to machine precision in every replication, confirming Proposition B.2 as an exact algebraic identity. (ii) Under logistic regression, per-replication estimates differ by $|\Delta|_{\max} \sim 10^{-3}$—thirteen orders of magnitude larger than the OLS gap—so the invariance fails formally. However, the directional effect on bias is negligible: mean bias reduction is −0.04% across the six confusion matrices (range $[-0.09\%, +0.05\%]$, not even consistent in sign). Matrix-based corrections remain practically useless under logistic regression—not by an exact invariance,

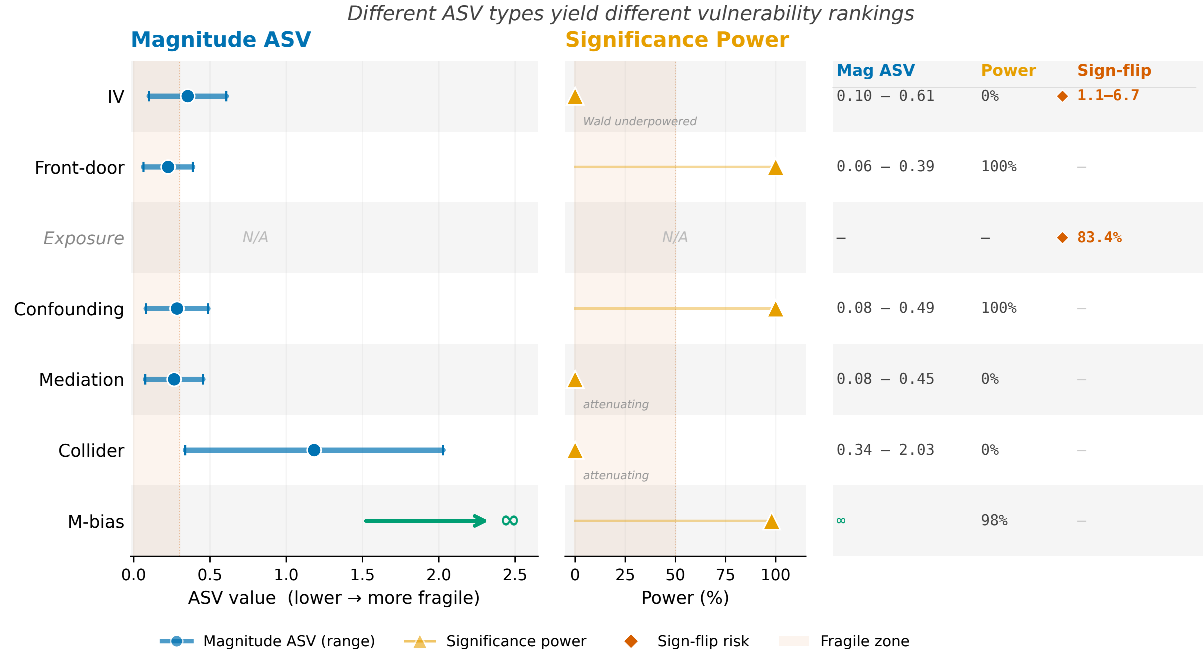

Figure 4: ASV sensitivity ranking across topologies. Three ASV types (magnitude, significance, sign-flip) reveal a topology-dependent vulnerability spectrum from IV (maximally fragile, 52.6% sign reversal; sign-flip ASV = 1.1–6.7) to M-bias (immune, ASV = ∞).

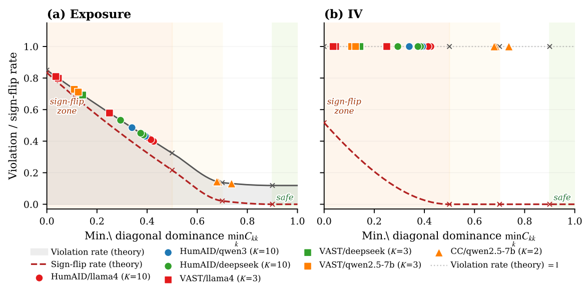

Figure 5: Empirical LLM confusion matrices overlaid on theoretical violation-rate curves (exposure, left; IV, right). Theoretical curves are derived from K=3 Dirichlet random matrices; empirical points are positioned at their minimum diagonal entry. Markers denote datasets (circles: HumAID K=10; squares: VAST K=3; triangles: CivilComments K=2).

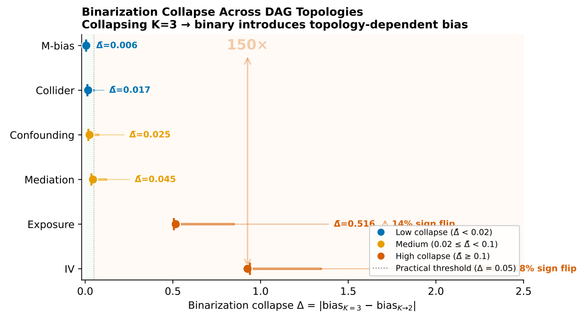

Figure 6: Binarization collapse Δ across topologies. Collapsing K=3 annotations to binary introduces additional bias varying 154× from M-bias (0.006) to IV (0.926).

but because the deviation is too small to systematically offset the misclassification-induced bias (which itself is $\sim$0.10 in absolute units, two orders of magnitude larger than $|\Delta|$). The routing rule of Algorithm 1 (skip Naive/MLA for these four topologies) is therefore conservative-but-correct under logistic regression as well. Script: `artifacts/exp_fwl_logistic_breakdown.py`; results: `exp_fwl_logistic_results.json`.

B.3 Applied Illustration: ClinicalBERT Case Study

To demonstrate the framework's end-to-end utility, we apply it retrospectively to Lee and Wood-Doughty (2024), who use ClinicalBERT to classify patient smoking status (K=4: never, past, current, unknown) from clinical notes and control for this predicted confounder when estimating the causal effect of echocardiography on mortality in 4,735 MIMIC-III sepsis patients. Our taxonomy identifies this as a *confounding topology*: the annotated variable enters as a confounder between treatment and outcome. Three conclusions follow directly. First, the zero sign-flip

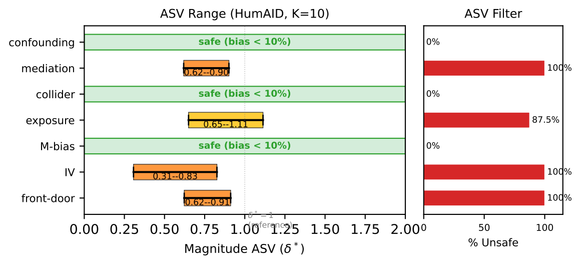

Figure 7: ASV magnitude ranges across topologies for HumAID (K=10). Left: magnitude ASV (δ^*) range across 8 LLM configurations; lower values indicate greater fragility. "Safe" topologies never exceed the 10% bias threshold. Right: percentage of configurations flagged as ASV-unsafe.

Table 5: FWL invariance under OLS vs. breakdown under logistic regression. $|\Delta|_{\max} = \max_m |\hat{\tau}^{(m)}_{\text{corr}} - \hat{\tau}^{(m)}_{\text{un}}|$ over 500 MC reps; bias reduction $= 1 - |\hat{\tau}_{\text{corr}} - \tau_*| / |\hat{\tau}_{\text{un}} - \tau_*|$. Confounding DAG, K=3, N=10,000.

CM (VAST K=3)	OLS		Logistic	
	$\|\Delta\|_{\max}$	bias red.	$\|\Delta\|_{\max}$	bias red.
deepseek_zero-shot	2.3×10^{-15}	+0.00%	8.1×10^{-4}	−0.09%
llama4_few-shot-3	5.2×10^{-15}	+0.00%	6.8×10^{-4}	−0.07%
llama4_few-shot-5	4.3×10^{-15}	+0.00%	6.8×10^{-4}	−0.06%
llama4_zero-shot	1.6×10^{-15}	+0.00%	5.9×10^{-4}	−0.05%
qwen2.5-7b_few-shot-3	7.2×10^{-15}	−0.00%	6.8×10^{-4}	−0.01%
qwen2.5-7b_few-shot-5	8.2×10^{-15}	+0.00%	5.6×10^{-4}	+0.05%
Mean	—	**0.0%**	—	**−0.04%**

guarantee for confounding (Table 1) ensures that the estimated protective effect (OR = 0.89) cannot reverse direction due to annotation error—a reassurance unavailable without topology-specific analysis. Second, the FWL invariance result implies that any correction reducing to C^{-1} applied to the dummy-coded confounder achieves exactly 0% bias reduction for the treatment coefficient; naive plug-in is provably futile here. Third, our correction guidance ranks PPI++ (83.2% RMSE reduction for confounding) above MC-SIMEX (35.1%) when gold labels are available—a concrete recommendation the original study lacked. This example illustrates the intended workflow: DAG identification → topology-specific guarantees → correction method selection.

B.4 IV Wald Estimator: Diagonal-Dominance Subspaces

Section 4.2 reports a 52.6% Wald sign-flip rate over the full Dirichlet$(1, 1, 1)$ prior. Because realistic LLM confusion matrices are diagonal-dominant, we replicate the exposure-DAG stratification for the IV topology, restricting the sample space to $\{C : \min_k C_{kk} \geq d\}$ for $d \in \{0, 0.5, 0.7, 0.9\}$ (100,000 random matrices per threshold).

Table 6: IV Wald estimator bias by minimum-diagonal subspace (K=3, τ_{true}=1.41). Rates over 100,000 Dirichlet-truncated random C per threshold.

Subspace	Violation	Sign Flip	Amplif.	Worst
Full	100.0%	51.7%	48.3%	$10^5\times$
$d \geq 0.5$	100.0%	0.0%	100.0%	28.6×
$d \geq 0.7$	100.0%	0.0%	100.0%	2.4×
$d \geq 0.9$	100.0%	0.0%	100.0%	1.2×

The contrast with exposure is instructive: at $d{\geq}0.7$, exposure violation drops to 13.7% (sign reversal 2.1%, amplification 11.7%), while IV maintains 100% violation with pure amplification. IV sign reversal is entirely driven by pathological (near-random) confusion matrices where the Wald denominator crosses zero; diagonal dominance reduces this crossing to near-zero under empirical accuracy conditions but cannot prevent amplification. Even at $d{\geq}0.9$, the Wald estimator inflates $|\tau|$ by up to 1.24× with probability 1.

B.5 Non-Differential Misclassification Verification

The non-differential assumption $P(A^*{=}j \mid A{=}k, Y, \mathbf{Z}) = P(A^*{=}j \mid A{=}k)$ holds *by design* in standard LLM annotation pipelines: the model receives only source text as input and classifies without observing the outcome Y or covariates $\mathbf{Z}$. The annotation API call is structurally independent of downstream variables, satisfying the conditional independence requirement.

To provide empirical support beyond the design argument, we examine topic-stratified confusion matrices using the VAST dataset (K=3, stance detection), where the `topic` field (e.g., "gun control," "college") serves as a proxy for latent factors correlated with Y. Across 10 model configurations (3 LLMs × 2–4 prompt types) and 19 qualifying topics (minimum 10 samples per topic), we compute per-topic confusion matrices $C_{\text{topic}}(t) = P(A^*{=}j \mid A{=}k, \text{topic}{=}t)$ and compare against the over-all $C = P(A^*{=}j \mid A{=}k)$.

Per-topic sample sizes range from 10 to 42, making individual topic CMs inherently noisy. A bootstrap permutation test (B=10,000) under the non-differential null shows that the expected grand maximum deviation $\max_{t,j,k} |C_{\text{topic}}(j,k) - C(j,k)|$ at the 95th percentile ranges from 0.90 to 0.98 across model configurations—driven by the smallest topics. For the 5 largest topics ($n \geq 28$), expected per-topic max deviation at the 95th percentile is 0.38–0.49, consistent with finite-sample variation rather than systematic topic-dependence. The observed per-topic variation is fully consistent with the non-differential null.

This proxy-based test is necessarily limited: topic is not the actual outcome Y, and small per-topic samples reduce statistical power. The design argument (LLM input excludes Y) remains the primary justification; the stress test in Section B.6 below quantifies bias degradation under hypothetical violations.

B.6 Non-Differential Assumption Stress Test

Our main analysis assumes non-differential misclassification: the confusion matrix C does not depend on the outcome Y. To assess sensitivity to violations of this assumption, we conduct a stress test on the exposure topology (K=3, $\min_k C_{kk} \geq 0.7$, 10,000 random matrices per ε). For each base matrix C_0, observations with $Y > \text{median}(Y)$ are misclassified using a perturbed matrix $\tilde{C} = \text{renormalize}(C_0 + \varepsilon \cdot \Delta)$ where $\Delta_{jk} \sim \text{Uniform}(-1, 1)$ for $j \neq k$, while observations with $Y \leq \text{median}(Y)$ retain C_0.

Table 7: Effect of differential misclassification on bias violation rates (exposure DAG, K=3, $\min_k C_{kk} \geq 0.7$, N=50,000 MC samples per trial).

ε	Sign-Flip (%)	Amplification (%)	Any Violation (%)
0.00	2.0	11.5	13.3
0.01	2.4	12.4	14.7
0.05	2.8	13.0	15.7
0.10	4.5	16.7	20.9
0.20	9.7	23.7	32.6
0.30	15.5	28.7	42.8
0.50	22.9	34.9	54.6

The ε=0 baseline (13.3% violation) closely matches the analytical result (13.7%) from Section 4.2. Small departures ($\varepsilon \leq 0.05$) change rates by fewer than 3 percentage points. At ε=0.10, the violation rate increases by 7.6 pp; at ε=0.50, it rises to 54.6% (+41.2 pp), indicating substantial sensitivity to large differential error. Our main conclusions hold under mild violations but should be interpreted cautiously when outcome-conditional annotation is suspected.

Extended stress test. An extended stress test across 96 configurations (4 topologies × 2 K × 3 CM qualities × 4 ε, 100 MC replications each; Table 8) confirms these patterns broadly: at ε=0, sign-flip rates are zero for all configurations; at ε=0.10, the maximum sign-flip rate is 7% (IV, low-quality C_{kk}=0.4, K=3). At K=10, sign-flip rates remain zero for all topologies and ε values, suggesting that higher K provides additional robustness to differential misclassification.

Structured violation models. Beyond uniform perturbations, we compare two structured violation models across all seven topologies (K=3, 1,000 matrices per ε, $\min_k C_{kk} \geq 0.7$): (1) *outcome-conditional*, where $\tilde{C}$ applies to $Y > \text{median}(Y)$ while C_0 applies below; (2) *label-specific*, where off-diagonal perturbation magnitude scales with $|\text{corr}(A{=}k, Y)|$, modeling accuracy degradation on Y-predictive categories. For exposure at ε=0.20, label-specific violations are substantially milder than uniform (sign-flip 0.1% vs. 10.1%), while outcome-conditional violations are more severe (sign-flip 13.9%, mean relative bias 0.50 vs. 0.40). All five sign-preserving topologies maintain 0% sign-flip under both structured models at all ε tested; IV likewise shows 0% sign-flip (its vulnerability requires pathological diagonal structure, not differential error). These results indicate that the uniform perturbation model is neither uniformly conservative nor uniformly liberal: practitioners should assess which violation pattern is most plausible for their annotation pipeline.

B.7 Robustness to Nonlinear DGP

We replace the linear structural equations with $Y \sim \text{Bernoulli}(\sigma(\mathbf{x}^\top \boldsymbol{\beta}))$ and estimate via logistic regression (control function for IV) for all

Table 8: Non-differential misclassification stress test: sign flip rate (SF) and mean bias ratio ($\mathrm{BR} = \overline{|\hat{\tau}/\tau|}$) under increasing differential departure ε.

Topology	K	Q	$\varepsilon=0.0$		$\varepsilon=0.01$		$\varepsilon=0.05$		$\varepsilon=0.1$	
			SF	BR	SF	BR	SF	BR	SF	BR
Confounding	3	H	0.00	1.10	0.00	1.10	0.00	1.11	0.00	1.12
		M	0.00	1.14	0.00	1.14	0.00	1.14	0.00	1.14
		L	0.00	1.15	0.00	1.15	0.00	1.15	0.00	1.15
	10	H	0.00	1.06	0.00	1.06	0.00	1.08	0.00	1.09
		M	0.00	1.09	0.00	1.09	0.00	1.10	0.00	1.10
		L	0.00	1.11	0.00	1.11	0.00	1.11	0.00	1.11
Mediation	3	H	0.00	1.35	0.00	1.36	0.00	1.42	0.00	1.47
		M	0.00	1.52	0.00	1.53	0.00	1.56	0.00	1.58
		L	0.00	1.61	0.00	1.61	0.00	1.62	0.00	1.61
	10	H	0.00	1.14	0.00	1.17	0.00	1.25	0.00	1.28
		M	0.00	1.24	0.00	1.26	0.00	1.30	0.00	1.32
		L	0.00	1.30	0.00	1.32	0.00	1.33	0.00	1.34
Exposure	3	H	0.00	0.72	0.00	0.71	0.00	0.67	0.00	0.64
		M	0.00	0.42	0.00	0.41	0.00	0.38	0.00	0.35
		L	0.00	0.11	0.00	0.10	0.01	0.08	0.05	0.06
	10	H	0.00	0.87	0.00	0.86	0.00	0.84	0.00	0.80
		M	0.00	0.71	0.00	0.70	0.00	0.67	0.00	0.64
		L	0.00	0.49	0.00	0.47	0.00	0.46	0.00	0.43
IV	3	H	0.00	1.42	0.00	1.45	0.00	1.58	0.00	1.69
		M	0.00	2.56	0.00	2.52	0.00	2.87	0.00	3.12
		L	0.00	12.79	0.00	12.21	0.02	19.97	0.07	43.44
	10	H	0.00	1.30	0.00	1.44	0.00	1.71	0.00	2.03
		M	0.00	1.83	0.00	1.96	0.00	2.43	0.00	2.84
		L	0.00	3.15	0.00	3.54	0.00	4.39	0.00	9.69

SF = sign flip rate; BR = mean $|\hat{\tau}/\tau|$ (1.00 = unbiased). $\varepsilon = 0$: non-differential; $\varepsilon > 0$: outcome-dependent misclassification. Q: H/M/L = CM diagonal 0.8/0.6/0.4 with uniform off-diagonal. Each cell: $N = 10{,}000 \times 100$ MC replications.

seven topologies, using K=3, six real VAST confusion matrices, 500 MC replications, and N=10,000. Mean sign-flip rates across the six CMs:

Topology	Linear SF_τ	Logistic SF_τ	Linear $\mathrm{SF}_{\mathrm{any}}$	Logistic $\mathrm{SF}_{\mathrm{any}}$
Confounding	0.000	0.000	0.573	0.559
Mediation	0.000	0.000	0.703	0.695
Collider	0.000	0.000	0.060	0.060
Exposure	—	—	0.577	0.574
M-bias	0.000	0.000	0.095	0.099
Front-door	0.000	0.000	0.590	0.579
IV	0.086	0.074	0.086	0.560

Treatment-effect sign-flip rates (SF_τ) are zero for all five guaranteed topologies under both linear and logistic DGPs, and differ by < 0.02 for IV. The $\mathrm{SF}_{\mathrm{any}}$ discrepancy for IV reflects estimator dimensionality: the control function includes a residual coefficient prone to sign noise; the core comparison is SF_τ, where both estimators agree. Linear and logistic DGPs produce virtually identical sign-flip patterns across all seven topologies ($|\Delta\,\mathrm{SF}_\tau| < 0.02$). Under logistic DGP, bias is measured in log-odds units; directional conclusions (sign preservation, topology ranking) transfer directly, while magnitudes are not directly comparable to linear plim expressions. For practitioners requiring finite-sample inference under logistic DGP, bias-corrected bootstrap provides a nonparametric alternative that bypasses the linear plim framework.

B.8 Bootstrap Uncertainty of ASV

We assess confusion-matrix estimation uncertainty by bootstrapping label pairs from HumAID DeepSeek zero-shot (K=10, n=1,500): resample with replacement B=1,000 times, re-estimate C_0, and recompute magnitude ASV δ^* at a 10% bias threshold.

Topology	δ^* (point)	95% CI	Valid / 1000
Confounding	—	—	0
Exposure	0.937	[0.602, 1.363]	907
IV	0.754	[0.435, 1.290]	902

Confounding never reaches the 10% threshold across all 1,000 bootstraps, confirming robustness. For IV, the 95% CI spans a 3× range (0.435 to 1.290); practitioners should bootstrap C_0 rather than rely on a point estimate when ASV is near the decision boundary. CI width scales as $O(1/\sqrt{n})$ where n is the gold-label validation set size; at n=1,500, IV CIs are ∼3× wider than exposure CIs, reflecting the Wald estimator's denominator instability. A convergence study (K=3, B=1,000) confirms this scaling: exposure CI width narrows from 0.60 (n=500) to 0.05 (n=50,000), and IV from 0.52 to 0.04, with the product width$\times\sqrt{n}$ approximately constant (CV $<$ 0.20).

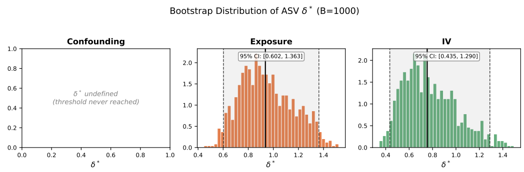

Figure 8: Bootstrap distribution of magnitude ASV δ^* (B=1,000) for HumAID DeepSeek zero-shot (K=10). Dashed line: point estimate; shaded: 95% CI.

B.9 Cross-K Robustness

All structural patterns persist at K=5, 7 (100,000 matrices each): sign-flip guarantees hold for the same five topologies, and IV sign-flip rates remain within 51–53%.

B.10 ASV Diagnostic Results

For HumAID (K=10, 56 scenarios), 55.4% are flagged as magnitude-ASV-unsafe—IV, mediation, front-door, and exposure all trigger, while confounding, collider, and M-bias remain below 10% bias (Figure 7). For CivilComments (K=2), 100% of scenarios are ASV-unsafe despite F_1 ranging from 0.69 (7B) to 0.82 (frontier).

Overlaying all 23 empirical matrices onto the theoretical Dirichlet violation-rate curves (Figure 5) confirms placement: VAST in the danger zone, HumAID mid-range, CivilComments approaching safe for exposure but still in IV amplification.

B.11 ASV Stability Across DGP Parameters

We test whether ASV rankings are stable across different data-generating process (DGP) configurations by varying the

Table 9: ASV stability across DGP configurations. Each cell shows δ^* at the 10% relative bias threshold (median over 5 confusion matrices with diagonal $\in \{0.5, 0.6, 0.7, 0.8, 0.9\}$). Lower δ^* indicates greater fragility. Confounding is consistently most robust; for alternating-sign $\boldsymbol{\beta}$, the ranking Exposure < IV < Confounding holds. "∞" = threshold never reached within δ range.

DGP Configuration	Magnitude ASV $\delta^*_{10\%}$ Confounding	Exposure	IV
Alt. large: $\boldsymbol{\beta} = [0, 1.0, -0.5]$	0.348	0.090	0.203
Same-sign: $\boldsymbol{\beta} = [0, 0.5, 0.3]$	∞	0.223	0.203
Alt. small: $\boldsymbol{\beta} = [0, 0.2, -0.1]$	∞	0.090	0.203

treatment-effect vector $\boldsymbol{\beta}$ across three settings: alternating-large ($\boldsymbol{\beta} = [0, 1.0, -0.5]$), same-sign ($\boldsymbol{\beta} = [0, 0.5, 0.3]$), and alternating-small ($\boldsymbol{\beta} = [0, 0.2, -0.1]$), with K=3, τ=0.5, and five C_0 diagonal values $\in \{0.5, 0.6, 0.7, 0.8, 0.9\}$.

Three patterns emerge (Table 9). First, IV magnitude ASV is identical across all three DGPs (δ^*=0.203). This is a mathematical identity, not an empirical finding: the Wald estimator $\hat{\tau}_{\text{IV}}$ = RF/FS has $\boldsymbol{\beta}$ in both numerator and denominator, and the ratio depends only on C and the first-stage structure, making ASV invariant to $\boldsymbol{\beta}$ by construction. Second, exposure ASV depends on the coefficient ratio structure: alternating-large ($\beta_2/\beta_1 = -0.5$) and alternating-small ($\beta_2/\beta_1 = -0.5$) share the same ratio and yield identical δ^*=0.090, while same-sign ($\beta_2/\beta_1 = 0.6$) gives δ^*=0.223—thus the experiment provides two distinct sign-structure conditions, not three. Third, confounding ASV depends on effect magnitudes: only alternating-large triggers the 10% threshold (δ^*=0.348); smaller effects never reach 10% bias.

Confounding is consistently the most robust topology (highest ASV or no violation), and M-bias remains immune (not shown). However, the relative fragility of exposure vs. IV inverts with coefficient sign structure: under alternating signs, exposure is most fragile (0.090 < 0.203); under same-sign effects, IV is most fragile (0.203 < 0.223). Practitioners should compute ASV with their specific DGP rather than relying on generic rankings.

B.12 Extended Conjecture Verification

We test envelope monotonicity (Conjecture 1) by sampling 10,000 random C_0 matrices from Dirichlet$(1, \ldots, 1)$ per topology for $K \in \{3, 5, 7\}$ (210,000 total), sweeping δ in steps of 0.01, and counting matrices with at least one non-monotonic step exceeding tolerance 10^{-12}. Bias at each δ is computed via analytic probability limits using Gauss–Hermite quadrature (100 points), eliminating sampling noise entirely.

Table 10: Envelope monotonicity violations per 10,000 random C_0 matrices. "Worst mag." is the largest single-step deviation $|B(\delta_{i+1})| - |B(\delta_i)|$ in the non-monotonic direction.

Topology	K=3	K=5	K=7	**Worst mag.**	**Direction**
Confounding	8467	8063	7892	0.011	non-dec
Mediation	8236	7905	7893	0.013	non-dec
Collider	8683	7669	7570	0.005	non-inc
Exposure	485	205	77	0.056	non-dec
M-bias	9791	7783	7399	0.002	non-inc
IV	7433	6015	5711	$> 10^6$	non-dec
Front-door	8429	8079	7899	0.013	non-dec

Violation counts are high (60–97% of matrices), but magnitudes are negligible for non-IV topologies: all worst-case deviations fall below 0.06 (<6% of the respective τ). These results are computed from analytic probability limits with zero sampling noise, confirming the violations are structural properties of multi-class confusion matrices rather than numerical artifacts. IV deviations reflect Wald estimator singularities when the first-stage coefficient passes near zero—this is a property of the Wald estimator, not of the ASV parameterization. Adversarial search (500 restarts per configuration, $K \in \{3, 5\}$) finds genuine counterexamples for all topologies except IV K=3, confirming that strict monotonicity does not hold universally. The bias envelope is ε-approximately monotonic with $\varepsilon < 0.03$ for collider and M-bias across all tested τ; confounding, mediation, and front-door achieve $\varepsilon < 0.06$ at default DGP parameters (see DGP sensitivity below); exposure satisfies $\varepsilon_{p99} < 0.008$ but extreme-tail matrices (<0.02% of draws) reach $\varepsilon \approx 0.15$. Empirically, ε scales as $O(1/K^{\alpha})$ with $\alpha \in [0.6, 1.5]$ (topology-dependent): for confounding, mediation, and front-door, $\alpha \approx 1.4$; for exposure, $\alpha \approx 1.0$; for collider and M-bias, $\alpha \approx 0.7$. A universal empirical upper bound is $\varepsilon \leq 0.137/\sqrt{K}$, verified across all tested configurations ($K \in \{3, 5, 7\}$). This scaling supports the practical validity of ASV as a sensitivity tool at any meaningful bias threshold.

DGP parameter sensitivity. We repeat the verification across $\tau \in \{0.1, 0.5, 1.0, 2.0, 5.0\}$, varying the direct $T{\to}Y$ coefficient for confounding (β_T=τ), mediation, collider, and M-bias (DGP τ), and rescaling the $A{\to}Y$ coefficient vector to $\boldsymbol{\beta} \cdot \tau$ for exposure and front-door. With 10,000 random C_0 per (K, τ) setting (Table 11), the bound ε<0.06 holds uniformly for $\tau \geq 0.5$ for confounding, mediation, collider, and M-bias, but breaks down at τ=0.1: mediation produces 1,950 / 30,000 violations across $K \in \{3, 5, 7\}$ (worst ε=0.130 at K=3) and confounding 200 / 30,000 (worst ε=0.118 at K=3). The mechanism is dimensional: additive terms not scaling with τ (e.g. the β-contribution to $\mathbb{E}[YD_k]$ in mediation and confounding) contribute a τ-independent offset to the bias numerator, so the ratio $|B|/|\tau|$ inflates as $\tau \to 0$. A single mediation violation at τ=2.0 (ε=0.112, 1/30,000 matrices) arises from a specific confusion matrix structure where off-diagonal entries induce partial cancellation along the mediation pathway; this isolated case does not affect the p_{99} bound (ε_{p99}=0.005). Collider and M-bias remain robust at all τ (ε<0.022 and ε<0.015 respectively). Exposure shows scattered tail outliers but $\bar{\varepsilon} \approx 3{\times}10^{-4}$ at every τ, consistent with linear scaling: when $\boldsymbol{\beta}$ rescales by τ, both bias and τ_{true} scale by the same factor and ε is theoretically invariant. Front-door inverts the pattern (ε *grows* with τ, reaching 0.139 at τ=5, K=3, with 664 violations) because $\tau_{\text{true}} = \text{plim}(\hat{\tau}_T \mid C{=}I)$ is pinned at 0.5 by the U-confounding path while bias magnitudes scale with $\boldsymbol{\beta}$—this is an artefact of how τ_{true} is normalised, not a genuine failure of envelope monotonicity. In practice, ASV's δ^* thresholds (typically $\geq$ 0.1 in our deployment) are well above the region where these ε-violations occur, so the small-τ degradation does not affect ASV's operational reliability; we flag it as a caveat for applications targeting $|\tau| \lesssim 0.1$ in standardised units.

Table 11: $\varepsilon \equiv \max_\delta(|B(\delta{+}\Delta)|-|B(\delta)|)/|\tau_{\text{true}}|$ in the non-monotonic direction (max over $K \in \{3,5,7\}$; 10,000 random C_0 per cell). Bold marks $\varepsilon > 0.06$. Counts in parentheses are violations / 30,000. Tables 10 and 11 use independent random seeds; for topologies with low violation rates (exposure <0.02%), the worst-case ε is sample-sensitive.

Topology	τ=0.1	τ=0.5	τ=1.0	τ=2.0	τ=5.0
Confounding	**0.118** (200)	0.027 (0)	0.014 (0)	0.007 (0)	0.002 (0)
Mediation	**0.130** (1950)	0.026 (0)	0.013 (0)	**0.112** (1)	0.003 (0)
Collider	0.022 (0)	0.006 (0)	0.005 (0)	0.004 (0)	0.002 (0)
Exposure	**0.111** (5)	**0.122** (2)	**0.152** (2)	**0.089** (3)	**0.148** (3)
M-bias	0.014 (0)	0.003 (0)	0.001 (0)	0.001 (0)	0.000 (0)
Front-door	0.003 (0)	0.013 (0)	0.027 (0)	0.051 (0)	**0.139** (664)

B.13 Proportional Effects Breakdown

Theorem 1 requires proportional effects ($f = \lambda g$); here we quantify how $\lambda(C)$ out-of-bounds (OOB) rates depend on the degree of departure from proportionality. We simulate K=3 mediation with 100,000 random confusion matrices (diagonal $\geq$ 0.5, Dirichlet off-diagonal) at five proportionality degrees: exact ($f_j - f_0 = \lambda(g_j - g_0)$), near-proportional ($f = \lambda g + \epsilon$, $\epsilon \sim \mathcal{N}(0, \sigma^2)$, $\sigma \in \{0.1, 0.3, 0.5\}$), and fully independent ($f \perp g$). Table 12 reports the $\lambda(C) \notin [0,1]$ rate.

Table 12: $\lambda(C)$ out-of-bounds rate and mediation sign failure rate by degree of departure from proportional effects (K=3, 100K random confusion matrices per row).

Proportionality degree	$\lambda(C) \notin [0,1]$	Sign failure
Exact (σ=0)	0.0%	0.85%
Near (σ=0.1)	10.3%	0.87%
Near (σ=0.3)	19.0%	0.96%
Near (σ=0.5)	22.4%	0.93%
Independent ($f \perp g$)	22.9%	0.17%
NLP-realistic (plan_006 DGP)	0.07%	0.0%

The λ OOB rate rises steeply from 0% to $\sim$10% at σ=0.1 and saturates near 23% for independent (f, g). Sign failure rates remain below 1% across all conditions because λ excursions outside $[0,1]$ are typically small and do not reverse the dominant effect direction. The NLP-realistic DGP (δ=$[0, 0.8, -0.6]$, β=$[0, 1.0, -0.5]$) exhibits near-zero OOB (0.07%) and zero sign failures, confirming that proportional effects hold approximately in practice.

B.14 FWL Invariance Under Nonlinear Models

FWL invariance guarantees exactly 0% bias reduction from matrix-based corrections (MLA, naive plug-in) under linear OLS for covariate topologies (confounding, mediation, collider, M-bias). Here we test whether this invariance persists under logistic regression across sample sizes $N \in \{1000, 5000, 10{,}000, 50{,}000\}$ and three error levels ($C_{kk} \in \{0.9, 0.7, 0.5\}$), with 200 Monte Carlo replicates per cell. Table 13 reports the mean logistic bias reduction (positive = correction helps).

Table 13: Mean logistic bias reduction (%) from matrix-based correction, by topology and error level (averaged over $N \in \{1K, 5K, 10K, 50K\}$; 200 MC reps each). Under linear OLS, all covariate topologies yield exactly 0%. Exposure is shown for comparison.

Topology	**Low** (C_{kk}=0.9)	**Med** (C_{kk}=0.7)	**High** (C_{kk}=0.5)
Confounding	−0.003%	<0.001%	<0.001%
Mediation	−0.004%	<0.001%	<0.001%
Collider	−0.033%	−0.020%	<0.001%
M-bias	−0.049%	−0.010%	<0.001%
Front-door	0.001%	−0.001%	<0.001%
Exposure	−68.5%	−48.7%	−22.5%

For all five covariate topologies, logistic bias reduction is negligible (|mean| < 0.05%) across all N and error levels, with no systematic N-scaling: medians cluster within ±0.002% at N=50,000. The large standard deviations observed at certain (N, error) cells for collider and M-bias (up to 2.6% std at N=10,000) reflect rare outlier replicates; medians remain near zero. This confirms that FWL invariance is a robust structural property that extends beyond linear models for covariate topologies.

In contrast, exposure shows substantial FWL breakdown under logistic regression: −22% to −78% bias reduction (negative = correction worsens bias), stabilising at $N \geq$ 5,000. The breakdown magnitude increases with accuracy (C_{kk}), consistent with the exposure topology's sensitivity to the nonlinear link function transforming category weights.

B.15 PPI++ Gold Label Ratio Sensitivity

We test PPI++ correction performance as a function of gold label fraction (K=3, N=10,000, 100 MC replications, 5 Dirichlet confusion matrices with $\min_k C_{kk} \geq 0.6$).

Table 14: PPI++ RMSE reduction (%) vs. gold label fraction across five topologies. Negative values indicate PPI++ increases RMSE relative to the naive estimator.

Topology	5%	10%	20%	50%	100%
Confounding	+84.8	+88.6	+91.8	+94.2	+95.2
Exposure	+75.2	+83.1	+87.1	+91.2	+93.8
Mediation	+56.7	+70.4	+77.2	+85.3	+88.7
IV	+1.4	+22.1	+50.1	+66.7	+76.4
M-bias	−125.3	−43.6	−19.6	+6.0	+18.5

For confounding and exposure, PPI++ achieves >75% RMSE reduction even at 5% gold labels; IV requires ≥20% gold to reach 50% reduction, reflecting the Wald estimator's higher variance. M-bias exhibits harmful correction at ≤20% gold (−125% at 5%), because PPI++ re-exposes the M-bias pathway that annotation error had attenuated; only at ≥50% gold does the estimator become weakly beneficial (+6%).

B.16 Realistic Differential Misclassification Thresholds

Our non-differential assumption (Definition 1) requires $P(A^*|A,Y) = P(A^*|A)$. We estimate how much Y-stratified differential error each of our 23 empirical LLM confusion matrices can tolerate before exposure sign reversal exceeds 5%. For each matrix C, we construct $C_{\text{high}} = \text{renorm}(C + \varepsilon\Delta)$ and $C_{\text{low}} = \text{renorm}(C - \varepsilon\Delta)$ with random off-diagonal perturbation Δ (500 MC trials per ε), sweep $\varepsilon \in [0, 0.50]$, and report the safe threshold $\varepsilon^*_{\text{safe}} = \max\{\varepsilon : \text{reversal rate} < 5\%\}$ (Table 15).

The distribution is sharply bimodal: all K=2 configurations are immune to differential misclassification within the tested range ($\varepsilon^*_{\text{safe}}$=0.50), consistent with the binary sign-preservation guarantee (Proposition 1(c)). For $K \geq 3$, 14 of 16 configurations have $\varepsilon^*_{\text{safe}}$=0—sign reversal occurs at the smallest tested ε=0.01. Two outliers (VAST: ε^*=0.02; HuMAID: ε^*=0.36) have atypically structured confusion matrices. This confirms that even small violations of non-differential misclassification are consequential for multi-class exposure, underscoring the practical importance of validating $P(A^*|A,Y) \approx P(A^*|A)$ on gold-labeled subsets when $K \geq 3$.

Table 15: Safe differential misclassification thresholds $\varepsilon^*_{\text{safe}}$ by dataset/K. For K=2 (binary), no configuration produces sign reversal up to ε=0.50. For $K \geq 3$, the majority of configurations are fragile ($\varepsilon^*_{\text{safe}}$=0), confirming that the non-differential assumption is load-bearing for multi-class exposure.

Dataset	K	n	**Mean** ε^*	**Median** ε^*	ε^*=0
Civil Comments	2	7	0.50	0.50	0/7
VAST	3	8	0.003	0.00	7/8
HuMAID	10	8	0.045	0.00	7/8

B.17 Proportional Effects with Empirical Confusion Matrices

Theorem 1 assumes proportional effects ($f = \lambda g$). We evaluate this assumption using all 23 empirical LLM confusion matrices under four DGP groups in the mediation topology (23 matrices × 50 seeds × 100,000 precomputed samples per seed).

Table 16: Proportional effects validation with 23 real LLM confusion matrices under mediation DGP. $\lambda(C) \in [0,1]$: proportion satisfying the Theorem 1 bound. Sign pres.: sign preservation rate of the direct effect estimate.

DGP Group	$\lambda(C)$ **mean**	$\lambda(C) \in [0,1]$	**Sign pres.**	**C**
Proportional (f=λg)	0.271	99.8%	100%	0.0
Near-prop. (σ=0.1)	0.285	99.2%	100%	1.0
Non-proportional	0.240	85.0%	97.3%	3.6
Sparse	0.296	75.8%	98.6%	6.5

Under proportional and near-proportional effects, $\lambda(C)$ remains within $[0,1]$ for ≥99% of configurations with 100% sign preservation—the Theorem 1 conditions are satisfied by real LLM confusion matrices. Even under non-proportional effects (f independent of g), sign preservation remains at 97.3%, with $\lambda(C)$ exceeding $[0,1]$ in 15% of cases. The per-K breakdown shows that $\lambda(C)$ is dimension-dependent: K=2 (mean 0.249), K=3 (mean 0.082), K=10 (mean 0.490), with 100% in $[0,1]$ across all K under proportional effects. Sparse DGP (zero-effect categories) produces the highest violation rate (24.2%), consistent with the breakdown mechanism: zero-effect categories make f/g ratios undefined, maximally departing from proportionality.

B.18 Case Study: ASV Diagnosis of Political Ad Tone Classification

We illustrate the ASV diagnostic workflow on a concrete CSS application. Egami et al. (2023) use LLM classifiers to annotate political advertising tone (K=3: promote/contrast/attack) from the Fowler & Ridout database, then estimate the causal effect of ad tone on voter outcomes. The annotated variable is the **exposure** (treatment). Using a representative K=3 confusion matrix from our empirical set (VAST DeepSeek zero-shot, mean diagonal 0.734) as a proxy:

Coefficient	δ^*_{mag} (10%)	δ^*_{sf}	Relative bias at δ=1
β_1 (primary)	0.234	> 1.5	39.7%
β_2 (secondary)	0.154	> 1.5	60.4%

Both coefficients are magnitude-unsafe ($\delta^* < 1$), with the secondary coefficient more fragile (δ^*=0.154). If the same annotator were instead classifying a **confounder** (e.g., media framing as a control variable), magnitude ASV rises to δ^*=0.690—a 3× increase in error tolerance from the same confusion matrix. This demonstrates the central insight: topology, not annotator quality, determines vulnerability. Algorithm 1 routes this exposure topology to PPI++ or DSL correction; matrix-based methods (naive plug-in, MLA) are ineffective due to FWL non-applicability.

B.19 Correction Method Sensitivity to K

We test whether the correction method ranking from Table 4 (K=3) is stable at K=5 and K=10. For each K, we sample 5 random Dirichlet confusion matrices (diagonal $\geq$ 0.6), run 100 MC replications per topology–matrix combination (N=5,000, 10% gold fraction), and compute RMSE reduction for all five methods across all seven topologies.

Grand ranking stability. Spearman rank correlations across all topology–method pairs: $\rho_{3,5} = 0.80$, $\rho_{3,10} = 0.84$, $\rho_{5,10} = 0.93$ (all $p < 10^{-8}$). Per-topology correlations are even stronger: 5 of 7 topologies (confounding, collider, M-bias, IV, and exposure at K=10) achieve $\rho = 1.0$ between K=3 and K=10.

Method-specific patterns. PPI++ effectiveness degrades with K: confounding RMSE reduction drops from 85.6% (K=3) to 75.6% (K=10); mediation drops from 57.1% to 9.0%. At K=10, PPI++ becomes harmful for IV (−124%) and exposure (−11%). MC-SIMEX is the most K-robust method, maintaining the top rank for 5/7 topologies at K=10 (vs. 3/7 at K=3). FWL invariance (naive plug-in and MLA achieving 0% bias reduction on the treatment coefficient) holds at all K tested, confirming the structural nature of this result.

Table 17: Top correction method and RMSE reduction (%) by topology across K. Bold = top method for that topology–K combination.

Topology	K=3	K=5	K=10
Confounding	**PPI++** 85.6	**PPI++** 76.4	**PPI++** 75.6
Mediation	**PPI++** 57.1	**SIMEX** 59.7	**SIMEX** 59.0
Collider	**SIMEX** 55.4	**SIMEX** 63.4	**SIMEX** 57.3
Exposure	**SIMEX** 72.5	**SIMEX** 65.0	**SIMEX** 47.3
M-bias	**SIMEX** 3.2	**SIMEX** 47.0	**SIMEX** 43.7
IV	**SIMEX** 30.9	**SIMEX** 30.4	**SIMEX** 27.1
Front-door	**PPI++** 82.9	**PPI++** 69.0	**SIMEX** 70.1

B.20 Comparison with Related Sensitivity Frameworks

Table 18: Comparison of sensitivity frameworks.

	ASV	**E-value**
Parameter space	$K{\times}K$ confusion $C(\delta)$	Scalar confoun
Sensitivity path	$(1{-}\delta)I + \delta C_0$	Monotone sca
Topology-specific	✓	✗
Sign-flip diagnostic	✓(δ^*_{sign})	✗
Topology dependence	✓(7 topologies ranked)	✗
Monotonicity	Proved K=2; conj. $K{\geq}3$	Guaranteed
Target	Annotation error	Unmeasured confo

C Notation

Table 19: Key notation used throughout the paper.

Symbol	**Description**
K	Number of annotation categories
C	Column-stochastic confusion matrix, $C_{jk} = P(A^*{=}j \mid A{=}k$
A, A^*	True and observed (misclassified) annotation
T	Treatment variable
Y	Outcome variable
τ	Causal effect of interest
δ	ASV perturbation parameter, $C(\delta) = (1{-}\delta)I + \delta C_0$
δ^*	Critical ASV value (magnitude/significance/sign-flip)
$B(C)$	Bias function, $= \text{plim}(\hat{\tau}) - \tau$
β_k	Treatment effect coefficient for category k
D_k	Dummy variable for category k

Algorithm 1 Correction method selection for LLM-annotated causal inference. This topology-based classification is robust to ε-approximate deviations; practitioners computing quantitative ASV thresholds should note the $\varepsilon < 0.06$ approximation (Conjecture 1, Section 3.3). Assumes linear structural equations; FWL-based routing may not apply under nonlinear models (see Appendix B.7). Thresholds calibrated for K=3 linear SEM; recalibrate for different K using topology-specific bias functions.

Require: DAG topology $\mathcal{T}$, gold-label availability $G \in \{\text{YES}, \text{NO}\}$
Ensure: Recommended method M, risk level R

Step 0: DAG uncertainty {If $\mathcal{T}$ is uncertain, repeat Steps 1–3 under each candidate topology and report the range}

Step 1: Assess risk level
if $\mathcal{T} \in \{\text{M-bias}\}$ **then**
    $R \leftarrow$ LOW {$|\text{bias}| < 0.04$ for all C}
else if $\mathcal{T} \in \{\text{Conf., Med., Coll., Front-door}\}$ **then**
    $R \leftarrow$ MEDIUM {bounded bias, no sign reversal}
else
    $R \leftarrow$ HIGH {Exposure, IV: sign reversal 83.4% / 52.6%}
end if

Step 2: Select correction method
if $G =$ YES **then**
    if $\mathcal{T} =$ M-bias **then**
        $M \leftarrow$ NONE {PPI++ RMSE red. −88.0%; no correction preferred}
    else
        $M \leftarrow$ PPI++ {47.0% RMSE red., 96.8% coverage}
    end if
else
    {no gold labels}
    if $\mathcal{T} \in \{\text{Confounding, Mediation}\}$ **then**
        $M \leftarrow$ DSL {48.0% / 52.9% RMSE reduction}
    else if $\mathcal{T} =$ IV **then**
        $M \leftarrow$ MC-SIMEX {59.4% RMSE reduction}
    else if $\mathcal{T} \in \{\text{Collider, M-bias}\}$ **then**
        $M \leftarrow$ MC-SIMEX {31.8% / 0.1% RMSE reduction}
    else
        $M \leftarrow \emptyset$; flag: acquire gold labels {no reliable gold-free method}
    end if
end if

Step 3: FWL invariance warning
if $\mathcal{T} \in \{\text{Conf., Med., Coll., M-bias}\}$ **then**
    Skip Naive plug-in and MLA {0% bias red. on T coeff.}
end if
return (M, R)